

Causal Inference for Unequal Baseline Panel Data Under Outcome Convergence

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Abstract

This paper considers causal inference in models where treatment and comparison groups have unequal treatment statuses in the pre-period, or “unequal baseline” treatment. A simple inversion of the Difference-in-Differences (DiD) model fails to identify treatment effects when treatment effects evolve over time, as is the case in many empirical settings. I explore an unequal baseline model with binary treatment and propose a causal inference approach that incorporates dynamic treatment effects. When level outcomes between Switchers and Stayers converge in the post-treatment period, treatment effects are estimable using simple methods and under mild assumptions. I prove that causal inference under this approach is valid for complex potential outcomes models such as interactive fixed effects. I target two parameters: the total treatment effect and the (instantaneous) extensive margin treatment effect. The event study estimation strategy I propose is easy to implement and provides an empirical test of the research design. I apply the methods derived in this paper to examine the role of the Pregnancy Discrimination Act of 1978 on women’s employment.

1 Introduction

Panel data models of treatment effects typically rely on a group unexposed to the treatment that serves as the control group. This control group enables the researcher to difference out any time invariant unobservables and common time trends between the two groups. Methods such as matching and synthetic controls allow flexibility in constructing these untreated groups, selecting units with similar pre-treatment distributions of the outcome, or with similar realizations of covariates. Despite advancement in these and other methods of selecting control groups, identifying untreated control groups is not always feasible in empirical applications for several reasons. For instance, in studying the effect of a federally-enacted policy, the researcher would not be able to identify any groups within the country that are not exposed to the treatment. In the case of targeted policies, selection criteria for treatment might prevent comparisons with untreated groups under canonical Difference-in-Differences (DiD) assumptions.¹ For example, it would be challenging to invoke canonical DiD assumptions in an examination of welfare policies that target populations experiencing a deterioration in outcomes relative to other groups. In these instances, panel data models often employ staggered treatment adoption designs, using groups that are not-yet-treated

¹If selection into treatment is correlated with pre-trends, traditional DiD assumptions fail in comparisons between treated and untreated groups.

as comparisons. Marcus and Sant’Anna (2021) and others have explored the tradeoffs in selecting not-yet-treated vs never-treated units as the control group, but the literature does not consider comparisons to units treated in the past. Furthermore, in the case of staggered treatment adoption over a wide period of time, early adopters’ treatment timing might be unknown. In some cases, the treatment timing for early adopters is known, but limited data availability may prevent us from observing outcomes prior to treatment. Constructing control groups of not-yet-treated units would require the researcher to exclude these groups, potentially exacerbating the efficiency losses identified in Marcus and Sant’Anna (2021).

When faced with this challenge of not being able to identify an untreated control group, researchers may proceed in a few ways. First, they might choose to redefine the model to examine the effect of a policy shock, rather than treatment. The group not experiencing a change in treatment is the comparison group and two-way fixed effects (TWFE) estimation methods such as Difference-in-Differences are used. When these estimands are mapped to potential outcomes, it becomes evident that the parallel trends assumptions required for identification deviate from the canonical form.² If treatment timing varies across groups, a staggered treatment adoption setting can be used. However, the standard approach requires omission of post-treatment periods in which all units are exposed to treatment. Depending on the length of time between treatment adoption across groups, this approach can often significantly restrict the length of time after treatment for which treatment effects can be identified, and in some cases, limits the number of treated units causing concerns for inference. An emerging literature presents an alternative approach in these settings: panel data models in which the treatment and control groups have unequal treatment statuses in the pre-period (Tazhitdinova and Vazquez-Bare 2023; K. Kim and Lee 2019). Instead of “treatment” and “control” groups, unequal baseline models characterize groups as Switchers and Stayers and invert the DiD model to identify treatment effects for the Switchers (Chaisemartin, D’Haultfœuille, and Vazquez-Bare 2024). However, these models rely on strong assumptions on the potential outcomes framework, treatment effect dynamics, and the evolution of outcomes for the Stayers. Because these restrictions are implausible in many empirical settings, an inversion of the simple DiD model often fails to recover causally interpretable coefficients.

In this paper, I study a subset of unequal baseline models in which causal inference is valid without significant restrictions on the evolution of treatment effects. When groups that are first exposed to treatment in different time periods exhibit convergent outcomes in the post-switch period, we can exploit time-variation in the outcome to identify treatment effects. In particular, time homogeneity restrictions on Stayers’ treatment effects are only necessary to impose when treatment affects Switchers and Stayers differentially. If outcomes converge across both groups as their treatment paths converge, their dimensions of heterogeneity in treated potential outcomes collapse. In other words, if post-switch period outcomes are equal on average up to a certain

²This approach identifies treatment effects when the time trend for the treated group unexposed to the shock is equal to the time trend for the treatment group that is exposed to the shock. If the treated group experiences treatment level d , the parallel trends assumption required for identification takes the form $\mathbb{E}[\Delta Y_t(d) \mid G = g] = \mathbb{E}[\Delta Y_t(0) \mid G = g']$.

degree of freedom, then it can be shown that their unobservable characteristics are also on average equal, up to the same degree of freedom. As long as treatment affects future outcomes for a finite number of periods, the observed fully-treated outcomes of the early-treatment group can model the counterfactual fully-treated outcomes of the Switching group. First-differencing between the observed untreated outcome and the counterfactual yields an estimate of the full treatment effect. This paper demonstrates that convergence in fully-treated outcomes is sufficient for identification of dynamic treatment effects even when treatment effects change over time, highlighting the robustness of the proposed estimator.

My approach demonstrates that it is possible to identify dynamic treatment effects that change both in event time (time relative to the unit’s first exposure to treatment) and calendar time. The research design allows identification of average treatment effects even when the true data generating process is complicated with time-varying unit-specific effects. I use an Interactive Fixed Effects (IFE) model for potential outcomes to demonstrate this feature, noting that as the number of post-treatment periods observed increases, the target parameter is identified for more and more complex potential outcomes models. Additionally, I differentiate between identification of total treatment effects, the effect of being exposed to treatment for a sufficient length of time, and extensive margin treatment effects, the instantaneous effect of exposure to treatment. This distinction is important when treatment effects are cumulative, and particularly when exposure to treatment has long-term effects on the outcome. Pinning down the exact treatment effect that is targeted in each estimation strategy meaningfully affects interpretation of the estimates. Both target parameters are meaningful in cost-benefit analysis and for welfare analysis, implying policy relevance of the target parameters identified in this paper.

The rest of the paper proceeds as follows: the remainder of section 1 summarizes the literature. Section 2 briefly introduces notation and formalizes the research design and identifying assumptions. In Section 2.3, I define the two target parameters. Section 3 is the main identification result, in which I show the sufficiency of Mean Outcome Convergence for balance across groups. In section 4, I propose event study estimators for the identified effects and derive their asymptotic properties. Section 5 generalizes results and extends the design to include covariates. Section 6 is an empirical illustration of the model that assesses the effects of the Pregnancy Discrimination Act of 1978 on employment, and Section 7 concludes.

Related Literature

This paper relates to three key areas in the causal inference literature. First is the literature on unequal baseline panel data models – settings in which the treatment and comparison groups experience different treatment status in the pre-period. Second is the growing literature on robust causal inference using multiple comparison groups, and lastly, the well-established literature on variation in treatment effect timing.

Recent work on panel data causal inference develop estimators for the unequal baseline setting. A Difference-in-Difference-in-Reverse (DDR) design uses an always-treated group as the control

group, making parallel trends assumptions about treated potential outcomes between Switchers and Stayers. This method can be used when treatment effects are non-cumulative, meaning that potential outcomes are invariant to the history of treatment (K. Kim and Lee 2019). However, it is empirically relevant to consider settings in which potential outcomes are dynamic with respect to the history of treatment. Tazhitdinova and Vazquez-Bare (2023) show that inference under canonical parallel trends is possible in this setting as long as the always-treated group does not experience dynamic treatment effects. In other words, treatment effects must be constant over time for the always-treated group. This restriction prohibits inference in settings where treatment effects are not homogeneous over time. They relax the non-cumulative effects condition with a linear trends adjustment, but note that it potentially exacerbates the bias if treatment effects accumulate non-linearly. Both of these papers impose that the Stayers behave like an untreated group in their identifying assumptions; my approach identifies treatment effects even when the control group is exposed to dynamic, cumulative treatment effects.

This paper is also closely related to the literature about selecting comparison groups. In an empirical study of state minimum wage policies, Dube (2019) highlights the importance of careful consideration when selecting control groups, emphasizing that Difference-in-Differences estimates are highly sensitive to comparison unit selection. Likewise, this paper proposes a testable research design that supports the plausibility of comparisons between the Switchers and Stayers. Callaway, Dyal, et al. (2025) establish an identification-strategy-robust approach to causal inference by selecting comparison groups that are “close” comparisons to the treatment group in the pre-treatment period. While their results apply to the equal baseline panel data model setting, my paper extends theirs by considering “close” comparisons between treatment groups and already-treated groups.

Lastly, this paper is related to the literature on forbidden comparisons. Many econometricians have evaluated the bias of TWFE estimation methods when treatment timing varies. Borusyak, Jaravel, and Spiess (2024) decompose the event study coefficients emphasizing that bias is attributed to both variation in treatment timing and treatment effect heterogeneity across groups. Relatedly, my approach isolates settings in which the latter converges across groups and exploits variation in the former to pin down treatment effects. See Baker (2019) for a summary of other seminal papers on TWFE bias. Goodman-Bacon (2021) evaluates a Difference-in-Differences approach when units experience treatment at different timings, and notes that because of treatment effect dynamics, comparisons between treated units and already treated units are forbidden comparisons in these settings. However, my approach demonstrates that in an unequal baseline setting, outcome convergence motivates the use of strictly forbidden comparisons in identifying treatment effects. In my setting, the negative weights from these comparisons are structurally relevant, particularly because universal adoption can be conceptualized as an inverse of the traditional DiD setting.

2 Identification

2.1 Setup

Consider a balanced panel with $i = 1, \dots, n$ units and T time periods $t = 1, 2, 3, \dots, T$. Treatments are binary, $d_t \in \{0, 1\}$, and potential outcomes are denoted as $Y_t(\mathbf{d}) = Y_t(d_1, \dots, d_{t-1}, d_t)$ where $\mathbf{d}$ is a t length vector of treatment assignments. Note that this setup allows for dynamic potential outcomes, where potential outcomes are a function of current treatment status as well as treatment history. It also implies a “no anticipation” assumption, that potential outcomes are agnostic to future treatment assignments. This is a standard assumption in panel data models and maintained throughout this paper. Units are grouped based on whether they switch treatment statuses over time. For simplicity, I define two groups with the switch occurring in $t = 3$, but results can later be generalized to multiple treatment groups. When treatment is binary, “unequal baseline” collapses to a universal adoption or policy reversal setting.

- **Universal Adoption:** The units that switch treatment status follow treatment path $\mathbf{d}^1 = (0, 0, 1, 1, \dots, 1)$ and are grouped as “Switchers” ($S=1$). The “non-switchers” or “always-treated” units follow treatment path $\mathbf{d}^0 = (1, 1, 1, \dots, 1)$ and are grouped as ($S=0$).
- **Policy Reversal:** In this setting, the non-switchers ($S=0$) are never treated, their treatment path is $\mathbf{d}^1 = (0, 0, 0, \dots, 0)$. Switchers are treated in the first two periods and become untreated in $t = 3$. Their treatment path is $\mathbf{d}^0 = (1, 1, 0, 0, \dots, 0)$

I focus discussion on the universal adoption setting. In the universal adoption setting, the comparison group is exposed to treatment and therefore experiences changes in the outcome from growth in untreated potential outcomes and from dynamic treatment effects. The necessary assumptions for identification in the universal adoption setting account for both sources of time trends. In the policy reversal case, the comparison group is never treated, meaning that dynamics for this group are mechanically restricted to changes over calendar time only. Therefore, identifying assumptions for the universal adoption setting are also sufficient for the policy reversal setting and some assumptions can be relaxed for the policy reversal setting.

2.1.1 Outcome Convergence Design

The model extrapolates on counterfactual outcomes for the Switchers in the pre-switch period relative to a group that, on average, experiences the same post-switch outcomes. Therefore, the setting requires that the Switchers and Stayers have equal observed outcomes in the post-switch period, after a finite transition period. Mathematically, the research design states for some $\tilde{t} \geq 3$,

$$\mathbb{E}[Y_t \mid S = 1] = \mathbb{E}[Y_t \mid S = 0] \quad \forall t \geq \tilde{t}$$

The research design is formalized in the following section.

2.1.2 Identifying Assumptions

Assumption 1: *Moment Conditions:* $(Y_{i1}, Y_{i2}, \dots, Y_{iT}; S_i)$ are *i.i.d* draws from a population and for all $t \in (1, T)$, $\mathbb{E}[Y_{it}] < \infty$, $\mathbb{E}[Y_{it}^2] < \infty$. Additionally, I assume that the number of Switchers and Stayers is positive, $\epsilon < \mathbb{E}[S] < 1 - \epsilon$ for some $\epsilon > 0$.

When there is variation in treatment timing, econometric models often require a structural assumption about carryover effects. For example, the “no carryover” or “negligible carryover effects” assumption is common in designs with variation in treatment timing (Arkhangelsky and Imbens 2024; Chaisemartin, D’Haultfœuille, and Vazquez-Bare 2024; Antonelli et al. 2025). Recent advancements consider the case of “limited carryover” effects, where treatment effects are allowed to evolve over time, but are restricted to a finite number of periods (Imai, I. S. Kim, and E. H. Wang 2023; Liu, Y. Wang, and Xu 2024). This paper builds on this literature by allowing for limited carryover effects in the context of unequal baseline treatment.

Assumption 2: *Limited Carryover:* Past treatment status affects potential outcomes for M periods with $M < \infty$.

$$Y_{it}(\mathbf{d}_{1:t}) = Y_{it}(\mathbf{d}_{t-M:t})$$

This assumption allows for treatment to affect future outcomes, and for treatment effects to evolve over time, but requires that potential outcomes do not have an infinite memory of past treatments. Intuitively, this means that once units have been treated for a sufficient number of periods, they can be considered “fully treated”, having reached their treated steady-state. This assumption is supported by economic theory when treatments are considered to have concave dynamic effects that diminish over time. Furthermore, limited carryover effects is empirically supported in many settings such as Fourniaies and Mutlu-Eren (2015).

To illustrate the main result, I will focus on a setting where $M = 1$. The limited carryover assumption requires that potential outcomes depend on current treatment status and treatment status in the preceding period only. Mathematically,

$$Y_{it}(\mathbf{d}_{1:t}) = Y_{it}(d_{t-1}, d_t).$$

Assumption 3: *Latent Unconfoundedness (Ω unobservable)*

$$Y_t(\mathbf{d}_{1:t}) \perp S \mid \Omega$$

This assumption states that conditional on a set of unobservables, Ω , potential outcomes are independent of treatment assignment. If Ω was observable, simple econometric methods would sufficiently recover causal parameters. Latent Unconfoundedness is sometimes referred to as Latent Ignorability in the literature (Liu 2026; Imai, I. S. Kim, and E. H. Wang 2023).

2.2 Research Design

I am interested in universal adoption settings in which, upon sufficient treatment history, average outcomes for the Switchers converge to the average outcomes of the Stayers. Formally, the Mean Outcome Convergence definition below must hold. Note that this is not an assumption, but rather an explicitly testable feature of the data.

Mean Outcome Convergence: *If the Switcher group becomes first treated at t^* , then for all time periods $t \geq t^* + M$,*

$$\mathbb{E}[Y_t(\vec{1}_{M+1}) \mid S = 0] = \mathbb{E}[Y_t(\vec{1}_{M+1}) \mid S = 1]$$

where $\vec{1}_{M+1}$ is an $M + 1$ length vector of ones, meaning that potential outcomes in this expression are with respect to full treatment status.

2.2.1 Potential Outcomes

The research design restricts the empirical settings in which this approach is applicable. However, beyond the empirically testable design, the model is highly flexible. A novel contribution of my approach is that it imposes fewer restrictions on the structure of potential outcomes, compared to the current DDR literature, by adopting a flexible factor model. Suppose the potential outcomes model has a factor structure with individual-level factor loadings, $\mathbf{W}_i \subseteq \Omega_i$ and time and treatment path-varying factors, $\mathbf{F}_{t,d_{t-M:t}}$. This allows the effect of individual unobservable factors to vary over time while still satisfying Assumption 3. Under the limited carryover assumption (Assumption 2), the potential outcomes model is,

$$Y_{it}(\mathbf{d}_{t-M:t}) = \beta_{i,t}(\mathbf{d}_{t-M:t}) + \mathbf{W}_i' \mathbf{F}_{t,d_{t-M:t}} + \epsilon_{it},$$

$$\mathbf{W}_i \subseteq \Omega_i$$

where $\beta_{i,t}(\mathbf{d}_{t-M:t})$ is the treatment effect for unit i at time t from treatment path $\mathbf{d}_{t-M:t}$. $\mathbf{W}_i' \mathbf{F}_{t,d_{t-M:t}}$ is the factor component that captures individual heterogeneity and time varying effects of heterogeneity on potential outcomes, and ϵ_{it} is idiosyncratic.

To account for time trends that are common across all units, the first elements of $\mathbf{W}_i$ and $\mathbf{F}_{t,d_{t-M:t}}$ are 1 and θ_t , respectively. To absorb any time-invariant, unit-specific effects, the last

elements of $\mathbf{W}_i$ and $\mathbf{F}_{t,d_{t-M:t}}$ are η_i and 1, respectively. For a model in which $\dim(\mathbf{W}_i) = 4$ ³,

$$\mathbf{W}_i = \begin{bmatrix} 1 \\ w_{1,i} \\ w_{2,i} \\ \eta_i \end{bmatrix} \text{ and } \mathbf{F}_{t,d_{t-M:t}} = \begin{bmatrix} \theta_t \\ f_{1,t} \\ f_{2,t} \\ 1 \end{bmatrix} \text{ for all } d_{t-M:t} \in \mathcal{D}.$$

Intuitively, this means that there are two distinct unobservables that affect potential outcomes differently over time while η_i continues to absorb unobservables that have the same effect over time, and θ_t continues to absorb common time trends. The appeal of the interactive fixed effects model in this setting is that with sufficiently high dimensional vectors $\mathbf{W}_i, \mathbf{F}_{t,d_{t-M:t}}$, it can approximate any outcome model (Bai 2009; Bai and Liao 2017; Athey et al. 2026). In practice, we are restricted by the number of post-treatment periods for which outcomes are observed; if the post-treatment period is very long, then this interactive fixed effects model can approximate more complicated outcome models. This is formally discussed in the main identification result section. For a model in which $\dim(\mathbf{W}_i) = K$,

$$\mathbf{W}_i = \begin{bmatrix} 1 \\ w_{1,i} \\ w_{2,i} \\ \vdots \\ w_{K-2,i} \\ \eta_i \end{bmatrix} \text{ and } \mathbf{F}_{t,d_{t-M:t}} = \begin{bmatrix} \theta_t \\ f_{1,t} \\ f_{2,t} \\ \vdots \\ f_{K-2,t} \\ 1 \end{bmatrix}.$$

My results will hold for the TWFE models in the current DDR literature because because the TWFE model is a special case of this interactive fixed effects equation where

$$\mathbf{W}_i = \begin{bmatrix} 1 \\ \eta_i \end{bmatrix} \text{ and } \mathbf{F}_{t,d_{t-M:t}} = \begin{bmatrix} \theta_t \\ 1 \end{bmatrix}.$$

Treatment effect heterogeneity

The treatment effect $\beta_{i,t}(\mathbf{d}_{t-M:t})$ can vary heterogeneously across individuals and across time. In the universal adoption setting, recovering treatment effects requires constructing the counterfactual *treated* potential outcome for the Switchers in the period before they switch. Standard settings require constructing counterfactual untreated potential outcomes. This distinction means that it is not sufficient to acknowledge that treatment effects are heterogeneous across groups and across

³I assume that $\mathbf{W}_i, \mathbf{F}_{t,d_{t-M:t}}$ are conformable for the dot product, so $\dim(\mathbf{F}_{t,d_{t-M:t}}) = \dim(\mathbf{W}_i)$. This must hold by construction because if $\mathbf{W}_i$ is of higher dimension than $\mathbf{F}_{t,d_{t-M:t}}$, then the individual factor loadings in $\mathbf{W}_i$ that map to the same time coefficients in $\mathbf{F}_{t,d_{t-M:t}}$ can be combined to reduce the dimension of $\mathbf{W}_i$. Conversely, if $\dim(\mathbf{W}_i) < \dim(\mathbf{F}_{t,d_{t-M:t}})$, then time coefficients can be similarly combined because of the distributive property.

units within groups, as this alone cannot be used to derive the counterfactual. Because of this, I model treatment effect heterogeneity as a function of a vector of the latent unobservables, $\mathbf{\Omega}$. Under this, $\beta_{i,t}(\cdot)$ can be written as a common function, $\beta_t(\cdot)$ of unit-level factors, $\mathbf{\Omega}_i$. For illustration, I model $\beta_t(\mathbf{\Omega}_i)$ with a factor model structure, noting that with sufficiently high dimension, this structure can approximate any model of treatment effect heterogeneity.

Corollary 1 to Assumption 3: $\mathbf{\Omega}$ is sufficient for treatment effect heterogeneity.

For all $t \in T$, for all i , and for all $M + 1$ length vectors of treatment paths, $\mathbf{d} \in \mathcal{D}$,

$$\beta_{i,t}(\mathbf{d}) = \mathbf{\Omega}_i' \mathbf{B}_{t,\mathbf{d}}$$

where $\mathbf{B}_{t,\mathbf{d}}$ is a vector of time and treatment path-varying coefficients.

Proof. By the law of iterated expectations,

$$\begin{aligned} \mathbb{E}[Y_{it}(\mathbf{d}_{t-M:t}) \mid S_i = s] &= \mathbb{E}[\mathbb{E}[Y_{it}(\mathbf{d}_{t-M:t}) \mid \mathbf{\Omega}_i, S_i = s] \mid S_i = s] \\ &= \mathbb{E}[Y_{it}(\mathbf{d}_{t-M:t}) \mid \mathbf{\Omega}_i] \\ &= \mathbb{E}[Y_t(\mathbf{d}_{t-M:t}) \mid \mathbf{\Omega}] \end{aligned}$$

The first equality holds by the law of iterated expectations, and the second equality follows from Assumption 3 and because the distribution of potential outcomes is independent of treatment group conditional on $\mathbf{\Omega}_i$. Therefore, $\mathbf{\Omega}$ is sufficient for potential outcomes. Because treatment effects are a function of potential outcomes (linear difference between treated and untreated potential outcomes), it follows that $\mathbf{\Omega}$ is also sufficient for treatment effect heterogeneity. $\square$

Remark 1: The factor model for untreated potential outcomes is a function of a subset of $\mathbf{\Omega}$. This allows for the possibility that treatment effect heterogeneity is a function of some unobservables that do not affect untreated potential outcomes. This is important because it models treatment effects that are heterogeneous across units even when untreated potential outcomes are homogeneous across units, contributing to the robustness of the identification strategy. In general, it is plausible to consider latent factor loadings that differentially affect treated potential outcomes but have no heterogeneous effects on untreated potential outcomes (e.g. latent skills that become salient only after the treatment is received). One can additionally consider the opposite: latent factor loadings that differentially affect baseline or untreated potential outcomes, but do not contribute to treatment effect heterogeneity. Then treatment effects can be modeled as a function of some subset of $\mathbf{\Omega}$ without loss of generality. This nuance is less conceptually likely in empirical settings and therefore not discussed. Other permutations are also feasible: for example, unobservables that affect untreated potential outcomes and unobservables that drive treatment effect heterogeneity can form two disjoint subsets of $\mathbf{\Omega}$. The rank condition on the Jacobian derivative matrix is with respect to the full set of latent unobservables, $\mathbf{\Omega}$. Therefore, the main identification result holds

for these and other variations of the model.

Assumption 4: *In the post-switch period, the Jacobian derivative matrix of fully-treated potential outcomes with respect to Ω is full rank.*

Let $\mathcal{G}_{\mathbf{d}}$ be the Jacobian derivative matrix of potential outcomes from treatment path $\mathbf{d} \in \mathcal{D}$ with respect to Ω . Additionally, assuming one-period carryover ($M = 1$), consider $t^* + 1 : T$ to span the post-switch period.

Following the factor structure, the rank condition on $\mathcal{G}_{1,1}$ can be shown by defining partition matrices $\mathbf{B}_{t,\mathbf{d}}^1$ and $\mathbf{B}_{t,\mathbf{d}}^2$ such that $\mathbf{B}_{t,\mathbf{d}} = [\mathbf{B}_{t,\mathbf{d}}^1, \mathbf{B}_{t,\mathbf{d}}^2]$ where $\mathbf{B}_{t,\mathbf{d}}^1$ is the submatrix of $\mathbf{B}_{t,\mathbf{d}}$ that maps to the first $\dim(\mathbf{W}_i)$ elements of Ω_i , and $\mathbf{B}_{t,\mathbf{d}}^2$ is the submatrix of $\mathbf{B}_{t,\mathbf{d}}$ that maps to the remaining elements of Ω_i . Then, the Jacobian derivative, $\mathcal{G}$, in the post-switch period can be written as a function of these partition matrices:

$$\mathcal{G}_{(1,1)} \Big|_{t^*+1:T} = \begin{bmatrix} \mathbf{G}_{t^*+1,(1,1)} \\ \mathbf{G}_{t^*+2,(1,1)} \\ \vdots \\ \mathbf{G}_{T,(1,1)} \end{bmatrix} = \begin{bmatrix} \mathbf{B}_{t^*+1,(1,1)}^1 + \mathbf{F}_{t^*+1,(1,1)} & \mathbf{B}_{t^*+1,(1,1)}^2 \\ \mathbf{B}_{t^*+2,(1,1)}^1 + \mathbf{F}_{t^*+2,(1,1)} & \mathbf{B}_{t^*+2,(1,1)}^2 \\ \vdots & \vdots \\ \mathbf{B}_{T,(1,1)}^1 + \mathbf{F}_{T,(1,1)} & \mathbf{B}_{T,(1,1)}^2 \end{bmatrix} \quad (1)$$

and must satisfy the condition: $\text{rank}(\mathcal{G}) = \dim(\Omega_i)$

This assumption requires that there is sufficient variation in outcomes in the post-switch period to identify the latent unobservables. Rank conditions on the Jacobian derivative matrix are standard in the literature on interactive fixed effects models (Bai 2009; Bai and Liao 2017; Athey et al. 2026). In this setting, it ensures that unit-level heterogeneity in the post-switch period can be extrapolated to pin down unit-level heterogeneity in the pre-switch period. Mathematically, it is that the factor structure in the pre-switch period is not of a higher dimension than the factor structure of the post-switch period. An example of a violation would be an unobservable (e.g. ability) that, across all units, affects potential outcomes in the pre-switch period but becomes irrelevant in the post-switch period.

2.3 Target Parameter

Focusing on the universal adoption setting with one carryover period, the key target parameters that I am interested in recovering are:

1. Total Average Treatment Effect (effect of being fully treated):

$$\begin{aligned} \forall t \leq t^* - 1, \\ \text{TATE}(s, t) = \mathbb{E}[Y_t(1, 1) - Y_t(0, 0) \mid S = s] \end{aligned}$$

The Total Treatment Effect on treated group s is the effect of reaching fully treated status,

meaning that in a setting with M carryover periods, it is the effect of being treated for at least $M + 1$ periods. I will focus the discussion on treatment effects in the period prior to the switch ($t = 2$). In this context, the targeted total treatment effect is:

$$\text{TATE}(1, 2) = \mathbb{E}[Y_2(1, 1) - Y_2(0, 0) \mid S = 1]$$

2. Extensive Margin Average Treatment Effect (Instantaneous treatment effect):

$$\begin{aligned} \forall t \leq t^* - 1, \\ \text{EMATE}(s, t) = \mathbb{E}[Y_t(0, 1) - Y_t(0, 0) \mid S = s] \end{aligned}$$

The Extensive Margin Average Treatment Effect on treated group s at time t is the isolated effect of being treated at time t . This can be conceptualized as the effect of “becoming treated” or the treatment effect on the extensive margin. Regardless of the number of carryover periods, it is the difference between potential outcomes with respect to being untreated in the past but treated at time t and untreated potential outcomes. In the model with one carryover period,

$$\text{EMATE}(1, 2) = \mathbb{E}[Y_2(0, 1) - Y_2(0, 0) \mid S = 1]$$

Both treatment effects are defined as the effect in $t = 2$ for the group that switches into treatment at $t = 3$, if they had been treated in $t = 2$. Thus, I discuss them as ATUs, or average treatment effects for the untreated. These target parameters have policy relevance as they are hypothetical treatment effects in early periods for later-treated groups. These parameters are instrumental to both cost effectiveness studies and welfare analyses of policies.

3 Identification

To build intuition, let $T = 5$ in the one-period carryover model, and suppose $\dim(\boldsymbol{\Omega}_i) = 2$.

The research design implies that we observe:

$$\begin{aligned} \mathbb{E}[Y_4(1, 1) \mid S = 0] &= \mathbb{E}[Y_4(1, 1) \mid S = 1] \\ \mathbb{E}[Y_5(1, 1) \mid S = 0] &= \mathbb{E}[Y_5(1, 1) \mid S = 1] \end{aligned}$$

Mapping these equalities to potential outcomes in the IFE model:

$$\begin{aligned}\mathbb{E}[\beta_{i,4}(1,1) + \mathbf{W}'_i \mathbf{F}_{4,(1,1)} \mid S = 0] &= \mathbb{E}[\beta_{i4}(1,1) + \mathbf{W}'_i \mathbf{F}_{4,(1,1)} \mid S = 1] \\ \mathbb{E}[\beta_{i,5}(1,1) + \mathbf{W}'_i \mathbf{F}_{5,(1,1)} \mid S = 0] &= \mathbb{E}[\beta_{i,5}(1,1) + \mathbf{W}'_i \mathbf{F}_{5,(1,1)} \mid S = 1]\end{aligned}\tag{2}$$

Notice that

$$\begin{aligned}\mathbb{E}[\beta_{i,t}(1,1) + \mathbf{W}'_i \mathbf{F}_{t,(1,1)} \mid S] &= \mathbb{E}[\boldsymbol{\Omega}'_i \mathbf{B}_{t,[1,1]} \mid S] + \mathbb{E}[\mathbf{W}'_i \mathbf{F}_{t,(1,1)} \mid S] \\ &= \mathbb{E}[\boldsymbol{\Omega}'_i \mid S] \mathbf{B}_{t,[1,1]} + \mathbb{E}[\mathbf{W}'_i \mid S] \mathbf{F}_{t,(1,1)}\end{aligned}\tag{3}$$

The first equality holds by applying the corollary to Assumption 3, that $\boldsymbol{\Omega}_i$ is sufficient for treatment effect heterogeneity, and the second expression holds because $\mathbf{B}_{t,[1,1]}$ and $\mathbf{F}_{t,(1,1)}$ are constant over units within each group and therefore constant across groups. The expectation of the dot product of a unit-specific vector with a constant vector is equal to the dot product of the expectations of the unit-specific vector and the constant vector. The last expression in Equation 3 implies that to show that treated potential outcomes are equal, on average, across groups, it is sufficient to show that the expected value of their unobservables, $\boldsymbol{\Omega}_i$, $\mathbf{W}_i$ are equal across groups. This sufficiency result leads to the identification of target parameters, and is formalized in the section that follows.

Proposition 1: Latent Variable Balance

Let $\boldsymbol{\Omega}_i$ be the vector of distinct factor loadings in the treated potential outcomes function. Under Assumptions 1-4, when Mean Outcome Convergence holds for $0 < K < T$ periods, and $\dim(\boldsymbol{\Omega}_i) = K$,

$$\mathbb{E}[\boldsymbol{\Omega}_i \mid S = 0] = \mathbb{E}[\boldsymbol{\Omega}_i \mid S = 1].$$

Proof. The IFE rank condition in Assumption 4 states that the Jacobian derivative of treated potential outcomes with respect to $\boldsymbol{\Omega}_i$ must be full rank. Notice that we can re-write the factor models in the potential outcomes framework by partitioning the matrices.

Partition $\beta_{i,t}$ in its factor structure model. $\beta_{it}(1,1) = \boldsymbol{\Omega}'_i \mathbf{B}_{t,[1,1]}$ with the following partitions,

$$\boldsymbol{\Omega}'_i \mathbf{B}_{t,(1,1)} = \begin{bmatrix} \mathbf{W}_i \\ \mathbf{W}_i^C \cap \boldsymbol{\Omega}_i \end{bmatrix}' \begin{bmatrix} \mathbf{B}_{t,(1,1)}^1 \\ \mathbf{B}_{t,(1,1)}^2 \end{bmatrix}.$$

Define the vector $\mathbf{G}_{t,(1,1)} = \begin{bmatrix} \mathbf{F}_{t,(1,1)} + \mathbf{B}_{t,(1,1)}^1 \\ \mathbf{B}_{t,(1,1)}^2 \end{bmatrix}$ Then, the factor model of $Y_{it}(1,1)$ can be written as:

$$\begin{aligned}
Y_{it}(1, 1) &= \mathbf{\Omega}'_i \mathbf{B}_{t,(1,1)} + \mathbf{W}'_i \mathbf{F}_{t,(1,1)} + \epsilon_{it} \\
&= \mathbf{\Omega}'_i \mathbf{G}_{t,(1,1)} + \epsilon_{it}
\end{aligned}$$

Under the research design, we observe

$$\begin{aligned}
\mathbb{E}[Y_{t^*+1}(1, 1) \mid S = 0] &= \mathbb{E}[Y_{t^*+1}(1, 1) \mid S = 1] \\
\mathbb{E}[Y_{t^*+2}(1, 1) \mid S = 0] &= \mathbb{E}[Y_{t^*+2}(1, 1) \mid S = 1] \\
&\vdots \\
\mathbb{E}[Y_T(1, 1) \mid S = 0] &= \mathbb{E}[Y_T(1, 1) \mid S = 1]
\end{aligned}$$

This implies that for K periods of mean outcome convergence, we have K equalities. For each $t \in \{t^* + 1, \dots, T\}$,

$$\begin{aligned}
\mathbb{E}[\mathbf{\Omega}'_i \mathbf{G}_{t,(1,1)} + \epsilon_{it} \mid S = 0] &= \mathbb{E}[\mathbf{\Omega}'_i \mathbf{G}_{t,(1,1)} + \epsilon_{it} \mid S = 1] \\
\mathbb{E}[\mathbf{\Omega}'_i \mathbf{G}_{t,(1,1)} \mid S = 0] &= \mathbb{E}[\mathbf{\Omega}'_i \mathbf{G}_{t,(1,1)} \mid S = 1]
\end{aligned}$$

The first line follows from the partition matrix representation of the factor model derived above and the second line follows from Assumption 1, since $\mathbb{E}[\epsilon_{it}] = 0$. Let $\mathcal{G}_{(1,1)}$ be the derivative matrix of fully-treated potential outcomes in the post-switch period, as defined in Equation 1. By Assumption 4, the matrix $\mathcal{G}_{(1,1)}$ has rank K . Note that this matrix is also square because we observe mean outcome convergence for K periods and $\dim(\mathbf{\Omega}_i) = K$. Therefore, $\mathcal{G}_{(1,1)}$ is invertible.

$$\begin{aligned}
\mathbb{E}[\mathbf{\Omega}'_i \mathcal{G}_{(1,1)} \mid S = 0] &= \mathbb{E}[\mathbf{\Omega}'_i \mathcal{G}_{(1,1)} \mid S = 1] \\
\rightarrow \mathbb{E}[\mathbf{\Omega}_i \mid S = 0]' \mathcal{G}_{(1,1)} &= \mathbb{E}[\mathbf{\Omega}_i \mid S = 1]' \mathcal{G}_{(1,1)} \\
\mathbb{E}[\mathbf{\Omega}_i \mid S = 0]' \mathcal{G}_{(1,1)} \mathcal{G}_{(1,1)}^{-1} &= \mathbb{E}[\mathbf{\Omega}_i \mid S = 1]' \mathcal{G}_{(1,1)} \mathcal{G}_{(1,1)}^{-1} \\
\mathbb{E}[\mathbf{\Omega}_i \mid S = 0]' \mathbf{I}_K &= \mathbb{E}[\mathbf{\Omega}_i \mid S = 1]' \mathbf{I}_K \\
\mathbb{E}[\mathbf{\Omega}_i \mid S = 0] &= \mathbb{E}[\mathbf{\Omega}_i \mid S = 1]
\end{aligned}$$

The first line expresses the system of K equalities as a dimension K matrix equality. The second follows because $\mathbb{E}[\mathcal{G}_{(1,1)} \mid S] = \mathbb{E}[\mathcal{G}_{(1,1)}] = \mathcal{G}_{(1,1)}$. The third line uses the fact that $\mathcal{G}_{(1,1)}$ is non-singular because it is a square matrix with full rank and right multiplies both sides of the equation by the inverse of $\mathcal{G}$. The fourth equality holds because $\mathcal{G}_{(1,1)} \mathcal{G}_{(1,1)}^{-1} = \mathbf{I}_K$ where $\mathbf{I}_K$ is a K -length identity matrix. The last expression is true because right-multiplying a $\dim(K)$ vector by $\mathbf{I}_K$ returns the original vector, and transposing both sides yields mean equality in $\mathbf{\Omega}$ across groups. $\square$

Note that this proof fails when $\dim(\mathbf{\Omega}_i) > K$ because the matrix $\mathcal{G}_{(1,1)}$ is no longer square and

therefore not invertible. In this case, the research design does not provide enough information to identify mean equality between the two groups in the latent factor space. The invertibility condition also fails when $\mathcal{G}_{(1,1)}$ is not full rank in the post-switch period because if $\mathcal{G}_{(1,1)}$ has a non-trivial null space, its determinant is zero.

Remark 2: The proof for Latent Variable Balance requires that the matrix of time-varying factors in the post-switch period has rank K , which is equal to the dimension of the vector of unobservable heterogeneity. Because $\dim(\boldsymbol{\Omega}_i)$ is unobserved by construction, the complexity of the model is restricted by the rank condition on $\mathcal{G}$. When more post-switch periods are observed, $\text{rank}(\mathcal{G})$ increases, and Proposition 1 holds for greater dimensions of heterogeneity. In DiD models, the literature indicates that observing outcomes in more pre-treatment periods can increase the statistical power of pre-trends tests. In contrast, in the setting of this paper, observing outcomes in more post-switch periods increases the structural validity of the potential outcomes model rather than contributing statistical power to estimation procedures.

Proposition 1(a)

If Mean Outcome Convergence holds for K periods, and $\dim(\boldsymbol{\Omega}_i) = K$, $\mathbb{E}[\boldsymbol{\Omega}_i \mid S = 0] = \mathbb{E}[\boldsymbol{\Omega}_i \mid S = 1]$ implies that for all $t \in T$,

$$\mathbb{E}[Y_t(1, 1) \mid S = 0] = \mathbb{E}[Y_t(1, 1) \mid S = 1]$$

In Proposition 1, I prove that the research design implies that the vector of latent variables that drive treatment effect heterogeneity is balanced across the two groups. Proposition 1(a) states that if the vector of latent variables is balanced across the two groups, then the average treated potential outcomes for the two groups are also equal in all time periods, not just in the post-switch period where this is observed to hold.

Proof. The proof of this proposition is straightforward.

Suppose Proposition 1 holds, so $\mathbb{E}[\boldsymbol{\Omega}_i \mid S = 0] = \mathbb{E}[\boldsymbol{\Omega}_i \mid S = 1]$.

For all $t \in T$,

$$\begin{aligned} Y_t(1, 1) &= \beta_{i,t}(\mathbf{d}_{t-\mathbf{M}}) + \mathbf{W}_i' \mathbf{F}_{t,(1,1)} + \epsilon_{it} \\ &= \boldsymbol{\Omega}_i' \mathbf{B}_{t,[1,1]} + \mathbf{W}_i' \mathbf{F}_{t,(1,1)} + \epsilon_{it} \\ &= \boldsymbol{\Omega}_i' \mathbf{G}_{t,(1,1)} + \epsilon_{it} \\ \mathbb{E}[Y_t(1, 1) \mid S = 0] &= \mathbb{E}[\boldsymbol{\Omega}_i \mid S = 0]' \mathbf{G}_{t,(1,1)} \end{aligned}$$

The first line uses the definition of potential outcomes, and the second line applies the factor model of treatment effect heterogeneity. The third line follows the partition matrix structure in the proof of proposition 1, and equality in the last line follows by taking conditional expectations of both

sides with respect to $S = 0$ and passing them through linearly. We can write the equality in the last expression in terms of the expected value of factor loadings for the Switchers.

$$\begin{aligned}\mathbb{E}[Y_t(1, 1) \mid S = 0] &= \mathbb{E}[\boldsymbol{\Omega}_i \mid S = 0]' \mathbf{G}_{t,(1,1)} \\ &= \mathbb{E}[\boldsymbol{\Omega}_i \mid S = 1]' \mathbf{G}_{t,(1,1)} \\ &= \mathbb{E}[Y_t(1, 1) \mid S = 1]\end{aligned}$$

The first line follows from proposition 1, that $\mathbb{E}[\boldsymbol{\Omega}_i \mid S = 0] = \mathbb{E}[\boldsymbol{\Omega}_i \mid S = 1]$ and the second line follows from the factor model for potential outcomes. $\square$

Remark 3: Because the factor loadings in the untreated potential outcomes function are a subset of the factor loadings for treatment effects, it might seem that this result also holds for $\mathbb{E}[Y_t(0, 0)]$. However, this is not necessarily the case because we do not impose rank conditions on the Jacobian derivatives $\mathcal{B}_{t,\mathbf{d}}$ and $\mathcal{F}_{t,\mathbf{d}}$ separately. Therefore, there is not enough information to conclude mean equality of untreated potential outcomes across groups. However, this is not a problem for identification of target parameters. In fact, I will highlight this as a strength of the model. If untreated potential outcomes diverge across cohorts while fully-treated potential outcomes converge, then it must be the case that treatment effects are heterogeneous across cohorts. Therefore, it follows that if $\mathcal{B}_{t,\mathbf{d}}$ or $\mathcal{F}_{t,\mathbf{d}}$ are not full rank, treatment effects are identified even when they are heterogeneous across groups.

3.1 Main Identification Result

I use the balance results from the above propositions in a decomposition of the target parameters. In the discussion that follows, I assume that treatment effects carry over for one period.

Proposition 2: *Under the research design and Assumptions 1-4, and when $M = 1$, the Total Treatment Effect is identified for all $t \leq t^* - 1$ and for all groups $s \neq 0$ as:*

$$\text{TATE}(s, t) = \mathbb{E}[Y_t \mid S = 0] - \mathbb{E}[Y_t \mid S = s]$$

Proof.

$$\begin{aligned}\text{TATE}(s, t) &= \mathbb{E}[Y_t(1, 1) - Y_t(0, 0) \mid S = s] \\ \text{TATE}(1, t) &= \mathbb{E}[Y_t(1, 1) \mid S = 1] - \mathbb{E}[Y_t(0, 0) \mid S = 1] \\ &= \mathbb{E}[Y_t(1, 1) \mid S = 0] - \mathbb{E}[Y_t(0, 0) \mid S = 1] \\ &= \mathbb{E}[Y_t \mid S = 0] - \mathbb{E}[Y_t \mid S = 1]\end{aligned}\tag{4}$$

The first line is the first target parameter defined in Section 2. The second line expresses the target

parameter in terms of the Switchers and passes expectations through linearly. The third line follows by using Proposition 1(a) because $\mathbb{E}[Y_t(1, 1) | S = 1] = \mathbb{E}[Y_t(1, 1) | S = 0]$. The last line expresses potential outcomes as observed outcomes because for the pre-switch period, all potential outcomes in line 3 are observed. $\square$

Extensive Margin Treatment Effect:

$$\begin{aligned}
\text{EMATE}(s, t) &= \mathbb{E}[Y_t(0, 1) - Y_t(0, 0) | S = s] \\
\text{EMATE}(1, t) &= \mathbb{E}[Y_t(0, 1) - Y_t(0, 0) | S = 1] \\
&= \underbrace{\mathbb{E}[Y_t(1, 1) - Y_t(0, 0) | S = 1]}_{\text{Total treatment effect}} - \underbrace{\mathbb{E}[Y_t(1, 1) - Y_t(0, 1) | S = 1]}_{\text{Lingering effect of past treatment}} \\
&= \mathbb{E}[Y_t(1, 1) | S = 0] - \mathbb{E}[Y_t(0, 0) | S = 1] - \underbrace{\mathbb{E}[Y_t(1, 1) - Y_t(0, 1) | S = 1]}_{\text{Lingering effect of past treatment}} \quad (5)
\end{aligned}$$

The first line is the target parameter defined in section 2 and the second line expresses the EMATE in terms of the Switchers at time t . The third line follows by adding and subtracting $\mathbb{E}[Y_t(1, 1) | S = 1]$. The fourth line uses Proposition 1(a) to substitute $\mathbb{E}[Y_t(1, 1) | S = 1] = \mathbb{E}[Y_t(1, 1) | S = 0]$. Intuitively, the total treatment effect at time t is the instantaneous effect of treatment at time t plus the lingering effect of having been treated for M periods in the past. It logically follows then that the EMATE can be expressed as a difference between the TATE and the lingering effect of past treatments.

A key challenge with this expression is that at $t \leq t^* - 1$, potential outcomes from becoming treated for the first time are not observed for any units. The next section will introduce additional assumptions on lingering effects of past treatment that are required to identify, or at least approximate, this effect.

3.1.1 Identifying Lingering Effects

Recall that the Switchers becomes treated for the first time at $t = 3$. This means that $\mathbb{E}[Y_3(0, 1) | S = 1]$ is observed, but given that treatment effects might be heterogeneous across time, this does not immediately identify $\mathbb{E}[Y_2(0, 1) | S = 1]$. Instead, we can approximate the lingering effects at $t = 2$ using a local linear approximation.

$$\mathbb{E}[Y_2(1, 1) - Y_2(0, 1) | S = 1] = \mathbb{E}[Y_3(1, 1) - Y_3(0, 1) | S = 1] - \nabla \mathbb{E}[Y_3(1, 1) - Y_3(0, 1) | S = 1] \quad (6)$$

Where ∇ is a backward difference operator, meaning $\nabla \mathbb{E}[Y_3(1, 1) - Y_3(0, 1) | S = 1]$ can alternatively be written as $\mathbb{E}[Y_3(1, 1) - Y_3(0, 1) | S = 1] - \mathbb{E}[Y_2(1, 1) - Y_2(0, 1) | S = 1]$. For notation simplicity, and to allow flexibility for a higher order approximation (such as a Taylor Series truncated to the second degree), I will maintain the ∇ notation to denote the backward derivative with respect to time.

Symmetric to the results in the previous sections, it can be shown that the first term is identified because redundancy implies $\mathbb{E}[Y_3(1,1) \mid S = 1] = \mathbb{E}[Y_3(1,1) \mid S = 0]$. Because $Y_t(0,1)$ is only observed at $t = 3$, we need additional assumptions on how lingering effects grow in order to identify the second term. Multiple approaches can lead to identification, and the approach will depend on the application. I provide two broad approaches and develop estimation under each. The first approach assumes that lingering treatment effects are constant over time and point identifies the target parameter. The second approach uses a bounded-variation approach on the growth rate of lingering treatment effects and leads to partial identification of $\text{EMATE}(s, t)$.

Assumption 5(a): *Stable lingering effects:*

The effect of being treated for M periods in the past is constant:

$$\begin{aligned} \forall t \leq T, \\ \nabla \mathbb{E}[Y_t(\vec{0}_M, 1) - Y_t(\vec{0}_{M+1}) \mid S] = 0 \end{aligned}$$

Equivalently,

$$\mathbb{E}[Y_t(\vec{1}_{M+1}) - Y_t(\vec{0}_M, 1) \mid S, t = t'] \text{ is invariant in } t'.$$

This equivalence follows by expanding the backwards difference operator and substituting the left hand side of the equation with $\nabla \mathbb{E}[Y_{t'}(\vec{0}_M, 1) - Y_{t'}(\vec{0}_{M+1}) \mid S]$.

Proposition 3: *Under Assumptions 1-4 and Assumption 5(a), $\text{EMATE}(s, t)$ is identified for all $j < t^* - 1$ and for all groups $s \neq 0$ as:*

$$\text{EMATE}(s, j) = \mathbb{E}[Y_j - Y_{t^*} \mid S = 0] - \mathbb{E}[Y_j - Y_{t^*} \mid S = s]$$

Proof. From the definition of the target parameter, and again assuming $M = 1$,

$$\begin{aligned} \text{EMATE}(s, t) &= \mathbb{E}[Y_t(0, 1) \mid S = s] - \mathbb{E}[Y_t(0, 0) \mid S = s] \\ &= \mathbb{E}[Y_t(1, 1) \mid S = s] - \mathbb{E}[Y_t(0, 0) \mid S = s] - \mathbb{E}[Y_t(1, 1) - Y_t(0, 1) \mid S = s] \end{aligned} \tag{7}$$

Notice that under Assumption 5(a),

$$\mathbb{E}[Y_t(1, 1) - Y_t(0, 1) \mid S = s] = \mathbb{E}[Y_{t^*}(1, 1) - Y_{t^*}(0, 1) \mid S = s]$$

This means that the expression in Equation 7 can be written:

$$\begin{aligned} \text{EMATE}(s, t) &= \mathbb{E}[Y_t(1, 1) \mid S = s] - \mathbb{E}[Y_t(0, 0) \mid S = s] - \mathbb{E}[Y_{t^*}(1, 1) - Y_{t^*}(0, 1) \mid S = s] \\ &= \mathbb{E}[Y_t(1, 1) \mid S = 0] - \mathbb{E}[Y_t(0, 0) \mid S = s] - \left(\mathbb{E}[Y_{t^*}(1, 1) \mid S = 0] - \mathbb{E}[Y_{t^*}(0, 1) \mid S = s] \right) \end{aligned} \quad (8)$$

The first equality comes directly from the last expression in equation 7 and the second equality holds by passing expectations through linearly in the second term and utilizing the result of Proposition 1(a). For all $j \leq t^* - 1$, untreated potential outcomes are observed for the Switchers and fully-treated potential outcomes are observed for the Stayers. Additionally, note that at $t = t^*$,

$$\mathbb{E}[Y_{t^*} \mid S = 0] = \mathbb{E}[Y_{t^*}(1, 1) \mid S = 0]$$

and

$$\mathbb{E}[Y_{t^*} \mid S = 1] = \mathbb{E}[Y_{t^*}(0, 1) \mid S = 1]$$

This means that all terms in the last line of Equation 8 are observed for time periods j such that $j \leq t^* - 1$, and the target parameter is identified as a function of observed outcomes:

$$\text{EMATE}(s, j) = \mathbb{E}[Y_j - Y_{t^*} \mid S = 0] - \mathbb{E}[Y_j - Y_{t^*} \mid S = s] \quad (9)$$

□

Remark 4: The extensive margin treatment effect is the effect of receiving treatment $d = 1$ for the first time, before outcomes evolve. For some treatments, the intensity of treatment itself is evolving in initial treatment periods. For example, a new law might require infrastructure changes or a transition period during which the stipulations of the law are not fully enforced. If this is the case, then the level of treatment in early periods is evolving, and $d_{t^*} < 1$ in periods immediately following the change. For the Switchers, $\mathbb{E}[Y_{t^*} \mid S = s] = \mathbb{E}[Y_{t^*}(\vec{0}_M, d_{t^*}) \mid S = s]$ for some $d_{t^*} < 1$. Differences in potential outcomes across groups are then driven both by differences in treatment history and by differences in treatment intensity. The estimated first difference might be driven by lingering treatment effects, but it also might be driven by the fact that Switchers experience a lower intensity of treatment, and it is impossible to separately identify the two effects. Researchers should consider this possibility and exercise caution when interpreting $\text{EMATE}(s, t)$ when analyzing treatments that carry this risk. However, in periods when treatment intensity is evolving, $d_t \in (0, 1]$. As long as cohort average treatment effects are monotonic in d_t , for any $d_t < 1$, the estimand yields a lower bound on the magnitude of treatment effects. Consequently, even if treatment intensity is evolving in initial switch periods, the approach yields a conservative estimate for $\text{EMATE}(s, t)$.

Although some settings, such as those with chemical treatments with a deterministic half life, may plausibly motivate a stable lingering effects assumption, others might not. The key difference

in my approach and that of K. Kim and Lee (2019) and Tazhitdinova and Vazquez-Bare (2023) is that my approach allows flexibility for dynamic treatment effects both in event time and calendar time. Therefore, it is worthwhile to allow this flexibility in not only the instantaneous treatment effect, but also the lingering treatment effect. Consider a technology-driven treatment for example. If technological advances improve the efficacy of treatment, then it is natural to think that these improvements manifest in both the effect of becoming treated and the lingering effects of past treatments. In this case, assumption 5(a) is not plausible.

Note that the equivalence expression in Assumption 5(a) can be equivalently stated as: $\nabla \mathbb{E}[Y_t(1, 1) | S] = \nabla \mathbb{E}[Y_t(0, 1) | S]$ which implies a common time trend between $Y_{it}(1, 1)$ and $Y_{it}(0, 1)$ within each group. Under Proposition 1(a), this is sufficient for parallel trends in treated potential outcomes across groups, and a 2x2 differencing estimand identifies the EMATE. If lingering treatment effects are time-variant, however, the TWFE estimate is biased, and the bias is equal in magnitude, and opposite in direction, to the growth in lingering treatment effects. In the general case, if lingering treatment effects are growing over time, then treatment effects identified under Assumption 5(a) are biased, and the bias at time t is equal to the difference in lingering treatment effects between time t and t^* .

Importantly, Assumption 5(a) is trivially satisfied when treatment effects are non-cumulative and non-dynamic, in which case it can be shown that $\mathbb{E}[Y_t(1, 1) - Y_{t-1}(1, 1) | S] = \mathbb{E}[Y_t(0, 0) - Y_{t-1}(0, 0) | S]$. However, when the time trend for fully treated potential outcomes deviates from the time trend for untreated potential outcomes, it is not clear whether potential outcomes from initiating treatment ($Y_t(0, 1)$) trend similar to untreated potential outcomes or treated potential outcomes. This is especially important if lingering treatment effects vary over time, in which case $Y_t(0, 1)$ certainly does not trend parallel to $Y_t(1, 1)$, but does not necessarily trend similar to $Y_t(0, 0)$ either. Instead of making strong and untestable assumptions about the growth rate, I exploit the difference in observed trends between untreated and fully treated potential outcomes to partially identify the treatment effect. Specifically, when growth trends in untreated and fully treated potential outcomes diverge substantially, the counterfactual time trend on $Y_t(0, 1)$ is bounded by the time trend on $Y_t(0, 0)$ and $Y_t(1, 1)$. Estimated growth rates based on $\nabla \mathbb{E}[Y_t(0, 0) | S]$ and $\mathbb{E}[Y_t(1, 1) | S]$ provide natural upper and lower bounds for the growth bias term.

To examine the deviation in time trends between $\mathbb{E}[Y_t(0, 0) | S]$ and $\mathbb{E}[Y_t(1, 1) | S]$, I employ methods similar to Rambachan and Roth (2023) and Manski and Pepper (2018), using the greatest observed difference in trends as an upper bound on this deviation. Let δ be the trend deviation between fully-treated outcomes of the Stayers and untreated outcomes of the Switchers. Using the factor structure maintained throughout this paper,

$$\begin{aligned}
\delta_t &= \nabla \mathbb{E}[Y_t(1, 1) - Y_t(0, 0) \mid S = s] = \mathbb{E}[Y_t(1, 1) - Y_{t-1}(1, 1) - (Y_t(0, 0) - Y_{t-1}(0, 0)) \mid S = s] \\
&= \mathbb{E}[Y_t(1, 1) - Y_{t-1}(1, 1) \mid S = 0] - \mathbb{E}[Y_t(0, 0) - Y_{t-1}(0, 0) \mid S = s] \\
&= (\mathbf{G}_t - \mathbf{G}_{t-1})' \mathbb{E}[\boldsymbol{\Omega}_i \mid S = 0] - (\mathbf{F}_t - \mathbf{F}_{t-1})' \mathbb{E}[\boldsymbol{\Omega}_i \mid S = s] \\
&= (\mathbf{G}_t - \mathbf{G}_{t-1} - [\mathbf{F}_t - \mathbf{F}_{t-1}])' \mathbb{E}[\boldsymbol{\Omega}_i \mid S = s]
\end{aligned} \tag{10}$$

The first line is the desired δ_t value, which is the second difference of fully treated and untreated potential outcomes over time. The second line uses Proposition 1(a) to write the expectation of fully treated potential outcomes for group s in terms of the Stayers, group 0. The third line maps the differences in observed outcome trends over time to their corresponding factor structures, and the fourth line uses the Latent Variable Balance result of Proposition 1 and the distributive property over $\mathbb{E}[\boldsymbol{\Omega}_i \mid S = s]$. δ_t is identified for all $t \leq t^* - 1$. Since this is the time period that we are conducting inference on, I am interested in characterizing the greatest deviation in trends between $\mathbf{G}_t$ and $\mathbf{F}_t$ for $t \leq t^* - 1$. Defining a vector of identified δ_t values, the greatest deviation in trends is identified in the pre-switch period as its infinity norm.

$$\boldsymbol{\delta} := \begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \\ \vdots \\ \delta_{t^*-1} \end{bmatrix}$$

And its infinity-norm,

$$\|\boldsymbol{\delta}\|_\infty = \max_{t \in [1, t^*-1]} |\delta_t| \tag{11}$$

Motivated by Rambachan and Roth (2023)'s approach, I use $\|\boldsymbol{\delta}\|_\infty$ to partially identify EMATE(s, t) under an assumption that lingering treatment effects are growing. The width of the interval is proportional to the divergence in time trends between $Y_t(0, 0)$ and $Y_t(1, 1)$, which is exactly $\|\boldsymbol{\delta}\|_\infty$. If $\|\boldsymbol{\delta}\|_\infty$ is large, then the treatment effect interval will also be wide. Conversely, if $\mathbb{E}[\nabla Y_t(1, 1) - \nabla Y_t(0, 0) \mid S = 1] = 0$, then EMATE(s, t) is point identified and equivalent to the naive approach. Therefore, when lingering treatment effects are varying over time, $\|\boldsymbol{\delta}\|_\infty$ is informative about the magnitude of bias from incorrectly assuming they are constant. Identification using a bounded-variation approach requires an assumption that variation over time in treatment effects are indeed bounded, as formalized below.

Assumption 5(b): *Trend Monotonicity*

The growth of first-treated potential outcomes with respect to time is bounded between the growth

of untreated and fully treated potential outcomes such that for all $t \in \mathcal{T}$:

$$\min \left\{ \nabla \mathbb{E}[Y_t(0,0) \mid S], \nabla \mathbb{E}[Y_t(1,1) \mid S] \right\} \leq \nabla \mathbb{E}[Y_t(0,1)] \leq \max \left\{ \nabla \mathbb{E}[Y_t(0,0) \mid S], \nabla \mathbb{E}[Y_t(1,1) \mid S] \right\}$$

This assumption states that, if treatment effects are positive, lingering treatment effects grow at most at the same rate as total treatment effects, and at least at a rate of 0. If treatment effects are negative, this assumption implies that lingering effects grow at a rate ranging from the growth rate of the total treatment effect to 0. Already, it is clear that one of the confidence bounds will be equivalent to the estimate under assumption 5(a); the 0 growth rate of lingering effects is equivalent to “stable lingering effects”, which is exactly assumption 5(a).

Proposition 4: *Under Assumptions 1-4 and Assumption 5(b), $EMATE(s,t)$ is partially identified for all $t < t^* + M$ and for all groups $s \neq 0$ and bounded between:*

$$\mathbb{E}[Y_{t^*} - Y_t \mid S = s] - \mathbb{E}[Y_{t^*} - Y_t \mid S = 0] \text{ and } \mathbb{E}[Y_{t^*} - Y_{t^*-1} \mid S = s] - \mathbb{E}[Y_{t^*} - Y_{t^*-1} \mid S = 0] + \|\delta\|_\infty.$$

Proof.

$$\begin{aligned} \mathbb{E}[Y_t(0,1) \mid S = s] &= \mathbb{E}[Y_{t^*}(0,1) \mid S = s] - \mathbb{E}[Y_{t^*}(0,1) - Y_t(0,1) \mid S = s] \\ &= \mathbb{E}[Y_{t^*}(0,1) \mid S = s] - (t^* - t) \cdot [\nabla \mathbb{E}[Y_r(0,1) \mid S = s]] \Big|_{r=t}^{t^*} \\ &\in \left[\mathbb{E}[Y_{t^*}(0,1) \mid S = s] - (t^* - t) \cdot [\nabla \mathbb{E}[Y_r(1,1) \mid S = s]] \Big|_{r=t}^{t^*}, \right. \\ &\quad \left. \mathbb{E}[Y_{t^*}(0,1) \mid S = s] - (t^* - t) \cdot [\nabla \mathbb{E}[Y_r(0,0) \mid S = s]] \Big|_{r=t}^{t^*} \right] \end{aligned} \quad (12)$$

The first equality adds and subtracts $\mathbb{E}[Y_{t^*}(0,1) \mid S = s]$. The second equality holds because using a simple linear expansion of the derivative, the difference in outcomes between two points in time is equal to the size of the interval multiplied by the growth rate. The third line applies assumption 5(b), which bounds the growth rate of $\mathbb{E}[Y_t(0,1) \mid S = s]$ between those of $\mathbb{E}[Y_t(0,0) \mid S = s]$ and $\mathbb{E}[Y_t(1,1) \mid S = s]$. This interval can be bound in terms of observed outcomes. Starting with the first bound in Equation 12:

$$\begin{aligned} \mathbb{E}[Y_{t^*}(0,1) \mid S = s] - (t^* - t) \cdot [\nabla \mathbb{E}[Y_r(1,1) \mid S = s]] \Big|_{r=t}^{t^*} \\ &= \mathbb{E}[Y_{t^*}(0,1) \mid S = s] - \mathbb{E}[Y_{t^*}(1,1) - Y_t(1,1) \mid S = s] \\ &= \mathbb{E}[Y_{t^*}(0,1) \mid S = s] - \mathbb{E}[Y_{t^*}(1,1) - Y_t(1,1) \mid S = 0] \\ &= \mathbb{E}[Y_{t^*} \mid S = s] - \mathbb{E}[Y_{t^*} - Y_t \mid S = 0] \end{aligned} \quad (13)$$

Which holds because the derivative over $[t, t^*]$ multiplied by the length of the interval, $t^* - t$, is equivalent to the linear difference in outcomes between these two points in time. The second line

uses Latent Variable Balance from Proposition 1(a) to write the second expectation over outcomes for $S = 0$. The third line holds because $\mathbb{E}[Y_{t^*}(0, 1) \mid S = s]$ is observed for Switchers, and $\mathbb{E}[Y_t(1, 1) \mid S = 0]$ is observed for all periods. Symmetrically, for the second bound in Equation 12:

$$\begin{aligned} \mathbb{E}[Y_{t^*}(0, 1) \mid S = s] - (t^* - t) \cdot [\nabla \mathbb{E}[Y_r(0, 0) \mid S = s]] \Big|_{r=t}^{t^*} \\ = \mathbb{E}[Y_{t^*}(0, 1) \mid S = s] - \mathbb{E}[Y_{t^*}(0, 0) - Y_t(0, 0) \mid S = s] \end{aligned} \quad (14)$$

The first term is observed in the data. Notice that for all $t \leq t^* - 1$, untreated potential outcomes are observed for the Switchers, meaning that the last term in the expression above is also observed. The challenge is that $\mathbb{E}[Y_{t^*}(0, 0) \mid S = s]$ is not observed for any groups. However, recall that $\mathbb{E}[Y_t(1, 1) \mid S = 0]$ is observed in all periods and furthermore recall that from Equation 11, the greatest divergence in trends is identified as $\|\delta\|_\infty$. Using the greatest pre-switch deviation in trends as an upper-bound and rearranging terms in Equation 10,

$$\begin{aligned} \mathbf{F}_{t^*} \mathbb{E}[\boldsymbol{\Omega}_i \mid S = s] &\leq (\mathbf{G}_{t^*} - \mathbf{G}_{t^*-1})' \mathbb{E}[\boldsymbol{\Omega}_i \mid S = 0] + \mathbf{F}_{t^*-1} \mathbb{E}[\boldsymbol{\Omega}_i \mid S = s] - \|\delta\|_\infty \\ &\leq \mathbb{E}[Y_{t^*}(1, 1) - Y_{t^*-1}(1, 1) \mid S = 0] + \mathbb{E}[Y_{t^*-1}(0, 0) \mid S = s] - \|\delta\|_\infty \\ &\leq \mathbb{E}[Y_{t^*} - Y_{t^*-1} \mid S = 0] + \mathbb{E}[Y_{t^*-1} \mid S = s] - \|\delta\|_\infty \end{aligned} \quad (15)$$

The first term takes that $\|\delta\|_\infty$ is the maximum deviation in trends in the period of interest and uses it as an upper-bound in trend deviations at t^* . The second line maps the factor structures to potential outcomes, and the third expression holds because these potential outcomes are observed. Since $\mathbb{E}[Y_{t^*}(0, 0) \mid S = s]$ enters the expression in 14 negatively, the identified expression above can be used to provide a conservative bound. The second bound on $\mathbb{E}[Y_t(0, 1) \mid S = s]$ can be written:

$$\begin{aligned} \mathbb{E}[Y_{t^*}(0, 1) \mid S = s] - \left(\mathbb{E}[Y_{t^*} - Y_{t^*-1} \mid S = 0] + \mathbb{E}[Y_{t^*-1} \mid S = s] - \|\delta\|_\infty - \mathbb{E}[Y_t(0, 0) \mid S = s] \right) \\ = \mathbb{E}[Y_{t^*} - Y_{t^*-1} \mid S = s] - \mathbb{E}[Y_{t^*} - Y_{t^*-1} \mid S = 0] + \|\delta\|_\infty + \mathbb{E}[Y_t \mid S = s] \end{aligned} \quad (16)$$

This holds because the first line replaces $\mathbb{E}[Y_{t^*}(0, 0) \mid S = s]$ in Equation 14 with the inequality derived in Expression 15 and the second line writes potential outcomes in terms of observed outcomes. The expressions in Equations 13 and 16 are fully identified bounds on $\mathbb{E}[Y_t(0, 1) \mid S = s]$, and can be used to partially identify the treatment effect. Notice that for all $t \leq t^* - 1$,

$$\begin{aligned}
\text{EMATE}(s, t) &= \mathbb{E}[Y_t(0, 1) - Y_t(0, 0) \mid S = s] \\
&= \mathbb{E}[Y_t(0, 1) \mid S = s] - \mathbb{E}[Y_t \mid S = s] \\
&\in \left[\mathbb{E}[Y_{t^*} - Y_t \mid S = s] - \mathbb{E}[Y_{t^*} - Y_t \mid S = 0], \right. \\
&\quad \left. \mathbb{E}[Y_{t^*} - Y_{t^*-1} \mid S = s] - \mathbb{E}[Y_{t^*} - Y_{t^*-1} \mid S = 0] + \|\delta\|_\infty \right]
\end{aligned}$$

The first equality is the definition of the target parameter, and the second expression follows because the potential outcome: $\mathbb{E}[Y_t(0, 0) \mid S = s]$ is observed during the pre-switch period. The third line substitutes the bounds from 13 and 16. $\square$

Under Assumption 5(b), the bounds on $\text{EMATE}(s, j)$ are generated by assuming that extensive margin treatment effects follow the same time trend as untreated potential outcomes (bound 2) or fully-treated potential outcomes (bound 1). It is worth understanding what each bound implies. If first-treated potential outcomes follow the same time trend as untreated potential outcomes, $\nabla \mathbb{E}[Y_t(0, 0) \mid S] = \nabla \mathbb{E}[Y_t(0, 1) \mid S]$, then instantaneous treatment effects are constant, and all differences in total treatment effects over time are attributed to the lingering treatment effect. This is unlikely since if treatment effects change over calendar time, we typically expect variation in the instantaneous treatment effect. Secondly, notice that the common trends assumption that generates bound 2: $\nabla \mathbb{E}[Y_t(1, 1) \mid S] = \nabla \mathbb{E}[Y_t(0, 1) \mid S]$ implies that all timing variation in total treatment effects can be attributed to the extensive margin effect, meaning that the time trend on lingering treatment effects is 0. This other extreme is exactly Assumption 5(a), and, as discussed above, is not supported by theory in many empirical settings. Because these two common trend assumptions span both extremes, assuming that treatment effects are bounded between them is a relatively mild assumption to impose, and this supports the feasibility of the trends monotonicity assumption in identification.

Several other approaches to estimating $\nabla \mathbb{E}[Y_t(1, 1) - Y_t(0, 1) \mid S = 1]$ are also feasible. For example, if $\mathbb{E}[Y_{t^*}(0, 0)]$ is projected using a local approximation method as discussed above, then a common proportions assumption on treatment effects: $\frac{\mathbb{E}[Y_t(0, 1) - Y_t(0, 0)]}{\mathbb{E}[Y_t(1, 1) - Y_t(0, 0)]} = \frac{\mathbb{E}[Y_{t+1}(0, 1) - Y_t(0, 0)]}{\mathbb{E}[Y_{t+1}(1, 1) - Y_t(0, 0)]}$ will point-identify treatment effects. This assumption would be feasible in settings where lingering effects and total treatment effects experience the same effects over calendar time. For instance, if treatment effects are constrained by technology limitations, and technology improves over time, we might expect both instantaneous and lingering effects of treatment to improve at the same rate. I do not consider the assumptions in this section to be exhaustive of those that are feasible and sufficient for identification, but rather a starting point.

4 Estimation

4.1 Total Treatment Effects

The population-level total treatment effect is identified in Proposition 2 as

$$\text{TATE}(s, t) = \mathbb{E}[Y_t \mid S = 0] - \mathbb{E}[Y_t \mid S = s].$$

Note that this is empirically equivalent to $-\mathbb{E}[(SS')^{-1}S'Y_t]$ which is the negative of the coefficient obtained from a linear regression of S on Y_2 . This expression indicates that the treatment effect can be estimated by taking the difference between the sample means of the outcome variable, across the two groups, at t :

$$\widehat{\text{TATE}}(1, 2) = \frac{1}{n_0} \sum_{i=1}^n Y_{i,2} \cdot (1 - S_i) - \frac{1}{n_1} \sum_{i=1}^n Y_{i,2} \cdot S_i$$

Where n_s is the number of observations in group s . $Y_{i,2}$ is the observed outcome for unit i at $t = 2$ and S_i is their treatment group ($S_i = 1$ for Switchers and $S_i = 0$ for Stayers).

For all pre-treatment periods,

$$\widehat{\text{TATE}}(s, t) = \frac{1}{\sum_i (1 - S_i)} \sum_{i=1}^n Y_{i,t} \cdot (1 - S_i) - \frac{1}{\sum_i S_i} \sum_{i=1}^n Y_{i,t} \cdot S_i \quad (17)$$

Where $Y_{i,t}$ is the observed outcome for unit i at time period t and $S_i = \mathbf{1}\{S_i = s\}$ is an indicator variable for treatment group membership. The first term is the sample mean of the outcome variable for the Stayers and the second term is the sample mean of the outcome variable for the Switchers.

This estimand can also be written as an influence function by combining the two sums and dividing by the proportional size of each group. This approach characterizes the estimand as a weighted average with weights according to the probability of treatment and the individual treatment assignment. However, in the setting of this paper, where mean outcome convergence is satisfied, the influence function approach does not add any insight and is therefore left off the estimation section. This approach is derived in Section 5 in the discussion about adding covariates, where it becomes clear that covariate balancing methods such as propensity score weighting are easy to implement, when necessary.

4.1.1 Asymptotic Distribution and Inference

Let $\text{Var}(Y_t \mid S = s) = \sigma_{s,t}^2$ for $s \in \{0, 1\}$ and $t \in \mathcal{T}$. Additionally, let $p = \Pr(S = 1)$ for notation convenience.

Proposition 5:

Under Assumptions 1-4, $\widehat{TATE}(s, t)$ is a consistent estimator of $TATE(s, t)$ and

$$\sqrt{n}(\widehat{TATE}(s, t) - TATE(s, t)) \xrightarrow{d} \mathcal{N}(0, \Sigma_t)$$

Where $\Sigma_t = \frac{p}{1-p} \mathbb{E}[Y_t | S = 0]^2 + \frac{1-p}{p} \mathbb{E}[Y_t | S = 1]^2 + \frac{\sigma_{0,t}^2}{1-p} + \frac{\sigma_{1,t}^2}{p}$.

Proof. By Assumption 1:

When units are i.i.d sampled from the entire population, S_i are i.i.d Bernoulli(p). Let the sample average be denoted: $\bar{S}_n = \frac{\sum S_i}{n} = \hat{p}$. By the weak law of large numbers, as $n \rightarrow \infty$,

$$\sqrt{n}(\hat{p} - p) \xrightarrow{d} \mathcal{N}(0, p(1-p)). \quad (18)$$

Because group membership is mutually exclusive, i.i.d random sampling of units also implies that $Y_{it} | S = s$ are independent draws from their respective conditional distributions, $Y_t | S = s$ for $s \in \{0, 1\}$. By the Weak Law of Large Numbers, as $n_s \rightarrow \infty$,

$$\bar{Y}_{tn_s} \xrightarrow{p} \mathbb{E}[Y_t | S = s] \quad (19)$$

Also note that the population-level total treatment effect can be expressed as a function of the conditional expectations of the outcome variable, given treatment group membership:

$$TATE(s, t) = \mathbb{E}[Y_t | S = 0] - \mathbb{E}[Y_t | S = s] = \frac{1}{1-p} \mathbb{E}[Y_t(1-S)] - \frac{1}{p} \mathbb{E}[Y_t S]$$

The pivotal quantity is:

$$\begin{aligned} \sqrt{n}(\widehat{TATE}(s, t) - TATE(s, t)) &= \sqrt{n} \left(\frac{1}{1-\hat{p}} \frac{1}{n} \sum_i Y_{it}(1-S_i) - \frac{1}{\hat{p}} \frac{1}{n} \sum_i Y_{it}S_i - \left(\frac{\mathbb{E}[Y_t(1-S)]}{1-p} - \frac{\mathbb{E}[Y_t S]}{p} \right) \right) \\ &= \sqrt{n} \left(\frac{1}{1-\hat{p}} \frac{1}{n} \sum_i Y_{it}(1-S_i) - \frac{1}{\hat{p}} \frac{1}{n} \sum_i Y_{it}S_i \right. \\ &\quad \left. - \left(\frac{1}{1-p} \frac{1}{n} \sum_i Y_{it}(1-S_i) - \frac{1}{p} \frac{1}{n} \sum_i Y_{it}S_i \right) \right. \\ &\quad \left. + \left(\frac{1}{1-p} \frac{1}{n} \sum_i Y_{it}(1-S_i) - \frac{1}{p} \frac{1}{n} \sum_i Y_{it}S_i \right) - \left(\frac{\mathbb{E}[Y_t(1-S)]}{1-p} - \frac{\mathbb{E}[Y_t S]}{p} \right) \right) \end{aligned}$$

The first equality restates the treatment effect as the population quantity as defined above and its sample analog. The second equality holds by adding and subtracting the same term. Notice that

this expression is an affine transformation of four distinct terms.

$$\mathbf{R}'\theta = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}' \begin{bmatrix} \sqrt{n} \cdot \left(\frac{1}{1-\hat{p}} - \frac{1}{1-p} \right) \frac{1}{n} \sum_i Y_{it}(1-S_i) \\ \sqrt{n} \cdot \left(\frac{1}{\hat{p}} - \frac{1}{p} \right) \frac{1}{n} \sum_i Y_{it}S_i \\ \sqrt{n} \cdot \frac{1}{1-p} \left(\frac{1}{n} \sum_i Y_{it}(1-S_i) - \mathbb{E}[Y_t(1-S)] \right) \\ \sqrt{n} \cdot \frac{1}{p} \left(\frac{1}{n} \sum_i Y_{it}S_i - \mathbb{E}[Y_tS] \right) \end{bmatrix} \quad (20)$$

Let $g(x) = \frac{1}{1-x}$ and $h(x) = \frac{1}{x}$. $g'(x) = \frac{1}{(1-x)^2}$ and $h'(x) = -\frac{1}{x^2}$. The asymptotic distribution of each element of θ is known.

$$\begin{aligned} \theta_1 &= \sqrt{n} \left(\frac{1}{1-\hat{p}} - \frac{1}{1-p} \right) \frac{1}{n} \sum_i Y_{it}(1-S_i) \\ &= \sqrt{n} (g(\hat{p}) - g(p)) \frac{1}{n} \sum_i Y_{it}(1-S_i) \end{aligned}$$

The first line is the first element of the θ vector and the second line applies the definition of $g(\cdot)$. Note that by the delta method, $\sqrt{n} (g(\hat{p}) - g(p)) \xrightarrow{d} N(0, g'(p)^2 \text{Var}(p))$ as $n \rightarrow \infty$. Combining this with the expression in 18, $\sqrt{n} (g(\hat{p}) - g(p)) \xrightarrow{d} N(0, \frac{p}{(1-p)^3})$. Additionally, from the Central Limit Theorem and Expression 19, $\frac{1}{n} \sum_i Y_{it}(1-S_i) \xrightarrow{p} \mathbb{E}[Y_t(1-S)] = (1-p)\mathbb{E}[Y_t | S=0]$.

By Slutsky's theorem,

$$\theta_1 \xrightarrow{d} \mathcal{N}\left(0, \frac{p}{(1-p)^3} \cdot ((1-p)\mathbb{E}[Y_t | S=0])^2\right) = \mathcal{N}\left(0, \frac{p}{1-p} \mathbb{E}[Y_t | S=0]^2\right).$$

Symmetrically,

$$\theta_2 = \sqrt{n} \left(\frac{1}{\hat{p}} - \frac{1}{p} \right) \frac{1}{n} \sum_i Y_{it}S_i \xrightarrow{d} \mathcal{N}\left(0, \frac{1-p}{p} \mathbb{E}[Y_t | S=1]^2\right).$$

For the third and fourth elements of θ , notice by the Central Limit Theorem,

$$\begin{aligned} \sqrt{n} \left(\frac{1}{n} \sum_i Y_{it}(1-S_i) - \mathbb{E}[Y_t(1-S)] \right) &\xrightarrow{d} \mathcal{N}(0, (1-p)\sigma_{0,t}^2) \\ \sqrt{n} \left(\frac{1}{n} \sum_i Y_{it}S_i - \mathbb{E}[Y_tS] \right) &\xrightarrow{d} \mathcal{N}(0, p\sigma_{1,t}^2) \end{aligned} \quad (21)$$

The asymptotic distributions of these elements are:

$$\theta_3 = \sqrt{n} \frac{1}{1-p} \left(\frac{1}{n} \sum_i Y_{it}(1-S_i) - \mathbb{E}[Y_t(1-S)] \right) \xrightarrow{d} \mathcal{N}\left(0, \frac{\sigma_{0,t}^2}{1-p}\right)$$

This follows because of the expression in 21 and because $\frac{1}{1-p}$ is a constant.

By symmetry,

$$\theta_4 = \sqrt{n} \frac{1}{p} \left(\frac{1}{n} \sum_i Y_{it} S_i - \mathbb{E}[Y_t S] \right) \xrightarrow{d} \mathcal{N} \left(0, \frac{\sigma_{1,t}^2}{p} \right)$$

Applying the continuous mapping theorem to Equation 20, as $n \rightarrow \infty$,

$$\sqrt{n} \left(\widehat{\text{TATE}}(s, t) - \text{TATE}(s, t) \right) = (\mathbf{R}' \theta) \xrightarrow{d} \mathcal{N} \left(0, \Sigma_t \right)$$

Where $\Sigma_t = \frac{p}{1-p} \mathbb{E}[Y_t | S = 0]^2 + \frac{1-p}{p} \mathbb{E}[Y_t | S = 1]^2 + \frac{\sigma_{0,t}^2}{1-p} + \frac{\sigma_{1,t}^2}{p}$. □

Variance Estimation

By Assumption 1, $\mathbb{E}[Y_t^2 | S = s] < \infty$ for $s = 0, 1$. Then by the Weak Law of Large Numbers, as $n_s \rightarrow \infty$,

$$\hat{\sigma}_{s,t}^2 = \frac{1}{n_s - 1} \sum_{i: S_i = s}^{n_s} (Y_{i,t} - \bar{Y}_t^s)^2 \xrightarrow{p} \sigma_{s,t}^2 \quad (22)$$

and $\sigma_{s,t}^2$ for $s \in \{0, 1\}$ can be consistently estimated by the sample variance of the outcome variable within each treatment group.

The variance of the estimator is

$$\text{Var}(\widehat{\text{TATE}}(s, t)) = \frac{1}{n} \left(\frac{p \mathbb{E}[Y_t | S = 0]^2 + \sigma_{0,t}^2}{1-p} + \frac{(1-p) \mathbb{E}[Y_t | S = 1]^2 + \sigma_{1,t}^2}{p} \right)$$

which follows directly from the expression in the limiting distribution. This can be consistently estimated by

$$\widehat{\text{Var}}(\widehat{\text{TATE}}(s, t)) = \frac{1}{n} \left(\frac{\hat{p} \left(\frac{1}{1-\hat{p}} \sum_{i=1}^n Y_{i,t} \cdot (1 - S_i) \right)^2 + \hat{\sigma}_{0,t}^2}{1 - \hat{p}} + \frac{(1 - \hat{p}) \left(\frac{1}{\hat{p}} \sum_{i=1}^n Y_{i,t} \cdot (1 - S_i) \right)^2 + \hat{\sigma}_{1,t}^2}{\hat{p}} \right). \quad (23)$$

4.2 Extensive Margin Treatment Effects

The precise estimation approach for extensive margin treatment effects depends on the identifying assumption on lingering effects assumed in Section 3.2.1. The identifying assumption determines the approach to approximating $\nabla \mathbb{E}[Y_{t+1}(1, 1) - Y_{t+1}(0, 1) | S = 1]$, the growth coefficient on lingering effects of past treatment.

4.2.1 Estimation under Assumption 5(a)

By proposition 3, the extensive margin treatment effect expression is:

$$\text{EMATE}(s, t) = \mathbb{E}[Y_t - Y_{t^*} \mid S = 0] - \mathbb{E}[Y_t - Y_{t^*} \mid S = s]$$

which can be approximated using its sample analog.

Event Study Specification

When the Stayers ($S_i = 0$) enter the model fully-treated, with 1 carryover period in potential outcomes, and when Switchers become first treated at $t = t^*$, the specification below identifies $\text{EMATE}(1, j)$ for all $j \leq t^* - 1$.

$$\widehat{\text{EMATE}}(s, j) = \frac{1}{\sum_i S_i} \sum_{i=1}^n (Y_{i,t^*} - Y_{i,j}) \cdot (S_i) - \frac{1}{\sum_i (1 - S_i)} \sum_{i=1}^n (Y_{i,t^*} - Y_{i,j}) \cdot (1 - S_i)$$

Where Y_{it} are the observed outcomes for unit i at time t , S_i is an indicator that equals 1 for Switchers. Note that the lowercase s is used in the event study to represent the specific switching group. The setting of this paper only considers one group that switches into treatment, but the estimator can be generalized to a case where there are multiple groups. When there are multiple Switcher groups, S_i is an indicator variable such that $S_i = \mathbf{1}\{S_i = s\}$.

4.2.2 Asymptotic Distribution and Inference

Let $\text{Var}(Y_t \mid S = s) = \sigma_{s,t}^2$ for $s \in \{0, 1\}$ and $t \in \mathcal{T}$. For notation simplicity, let $p = \text{Pr}(S = 1)$ and $\Sigma_t^2 = \frac{\sigma_{0,t}^2}{1-p} + \frac{\sigma_{1,t}^2}{p}$ as derived in Proposition 5.

Proposition 6:

Under Assumptions 1-4 and Assumption 5(a), for all $j \leq t^ - M$, and for all groups $s \neq 0$*

$$\widehat{\text{EMATE}}(s, j) \xrightarrow{d} \mathcal{N}\left(\text{EMATE}(s, j), \frac{1}{n} \mathbf{V}\right) \text{ as } n \rightarrow \infty.$$

where $\mathbf{V} = \Sigma_{t^*}^2 + \Sigma_j^2 - 2 \left(\frac{\text{Cov}(Y_{t^*}, Y_j \mid S=0)}{1-p} + \frac{\text{Cov}(Y_{t^*}, Y_j \mid S=s)}{p} \right)$

Proof. Under Assumption 1, the event study estimator decomposes as follows:

$$\begin{aligned}
\widehat{\text{EMATE}}(s, j) &= \frac{\sum_{i=1}^n (Y_{it^*} - Y_{ij}) \cdot S_i}{\sum_{i=1}^n S_i} - \frac{\sum_{i=1}^n (Y_{it^*} - Y_{ij}) \cdot (1 - S_i)}{\sum_{i=1}^n (1 - S_i)} \\
&= \frac{\sum_{i=1}^n (Y_{it^*} \cdot S_i)}{\sum_{i=1}^n S_i} - \frac{\sum_{i=1}^n Y_{it^*} \cdot (1 - S_i)}{\sum_{i=1}^n (1 - S_i)} - \left(\frac{\sum_{i=1}^n (Y_{ij} \cdot S_i)}{\sum_{i=1}^n S_i} - \frac{\sum_{i=1}^n Y_{ij} \cdot (1 - S_i)}{\sum_{i=1}^n (1 - S_i)} \right) \\
&= \bar{Y}_{t^*}^1 - \bar{Y}_{t^*}^0 - \left(\bar{Y}_j^1 - \bar{Y}_j^0 \right)
\end{aligned}$$

The first equality is the event study estimator of $\text{EMATE}(s, j)$, which is the difference in outcomes at time j between the target group and the comparison group, relative to their differences in outcomes at the reference time period, t^* . The second line follows by distributing the summations, and the third line expresses plug-in sample average estimates as $\bar{Y}_t^s$ for $s = 0, 1$.

By Proposition 5, for all $t \in [0, \mathcal{T}]$, the asymptotic distribution of the first difference across groups is normal:

$$\sqrt{n} \left(\bar{Y}_t^0 - \bar{Y}_t^1 - (\mathbb{E}[Y_t | S = 0] - \mathbb{E}[Y_t | S = 1]) \right) \xrightarrow{d} \mathcal{N}(0, \Sigma_t).$$

Additionally, because $\text{TATE}(s, t)$ is the first difference estimator between group s and group 0 at time t , $\text{EMATE}(s, j)$ is a linear transformation of $\text{TATE}(s, j)$ and $\text{TATE}(s, t^*)$.

By the bivariate Cramer-Wold theorem,

$$\sqrt{n} \left(\widehat{\text{TATE}}(s, t^*) - \text{TATE}(s, t^*), \widehat{\text{TATE}}(s, j) - \text{TATE}(s, j) \right) \xrightarrow{d} N(0, \mathbf{D}),$$

where $\mathbf{D}$ is a 2x2 covariance matrix.

$$\mathbf{D} = \begin{bmatrix} \Sigma_{t^*} & \frac{1}{p} \text{Cov}(Y_{t^*}^*, Y_j | S = s) + \frac{1}{1-p} \text{Cov}(Y_{t^*}^*, Y_j | S = 0) \\ \frac{1}{p} \text{Cov}(Y_{t^*}^*, Y_j | S = s) + \frac{1}{1-p} \text{Cov}(Y_{t^*}^*, Y_j | S = 0) & \Sigma_j \end{bmatrix}$$

Which follows because

$$\text{Cov}(\widehat{\text{TATE}}(s, t^*), \widehat{\text{TATE}}(s, j)) = \frac{\text{Cov}(Y_{t^*}^*, Y_j | S = 0)}{1 - p} + \frac{\text{Cov}(Y_{t^*}^*, Y_j | S = 1)}{p}.$$

This equation holds because under Assumption 1, outcomes are independent across units, and the covariance between the two estimators is a function of the covariance between the outcomes at time t^* and j within each group.

Since the difference of the two estimators is a continuous linear transformation, the Continuous Mapping Theorem states:

$$\sqrt{n} \left(\widehat{\text{TATE}}(s, t^*) - \widehat{\text{TATE}}(s, j) - [\text{TATE}(s, t^*) - \text{TATE}(s, j)] \right) \xrightarrow{d} N(0, \mathbf{V})$$

Where

$$\begin{aligned}\mathbf{V} &= \text{AVar}\left(\widehat{\text{TATE}}(s, t^*)\right) + \text{AVar}\left(\widehat{\text{TATE}}(s, j)\right) - 2\text{Cov}\left(\widehat{\text{TATE}}(s, t^*), \widehat{\text{TATE}}(s, j)\right) \\ &= \Sigma_{t^*} + \Sigma_j - 2\left(\frac{\text{Cov}(Y_{t^*}, Y_j \mid S = 0)}{1 - p} + \frac{\text{Cov}(Y_{t^*}, Y_j \mid S = s)}{p}\right).\end{aligned}$$

□

Variance Estimation Variance estimation for the extensive margin treatment effect estimator is straightforward. The variance of the estimator follows directly from the limiting distribution in Proposition 6.

$$\begin{aligned}\text{Var}(\widehat{\text{EMATE}}(s, j)) &= \frac{p(\mathbb{E}[Y_{t^*} \mid S = 0]^2 + \mathbb{E}[Y_j \mid S = 0]^2) + \sigma_{0,t^*}^2 + \sigma_{0,j}^2 - 2\text{Cov}(Y_{t^*}, Y_j \mid S = 0)}{1 - p} \\ &\quad + \frac{(1 - p)(\mathbb{E}[Y_{t^*} \mid S = s]^2 + \mathbb{E}[Y_j \mid S = s]^2) + \sigma_{s,t^*}^2 + \sigma_{s,j}^2 - 2\text{Cov}(Y_{t^*}, Y_j \mid S = s)}{p}.\end{aligned}$$

This can be estimated using sample analogs and the bias-corrected sample variance estimator in equation 22.

The proposed event study estimators for $\text{TATE}(s, t)$ and $\text{EMATE}(s, t)$ are point-wise estimators for each pre-switch period, t . These estimators are operationalization outside of the linear regression. I discuss the relationship between the proposed estimators and event study regressions in Appendix A.

4.3 Identification Test

So far, I have taken as given that Mean Outcome Convergence holds. While this is a strong condition, I also argue that it is a testable feature of the data rather than an assumption. In this section, I consider empirical tests of the research design.

For a given carryover period M , the difference in means in the sample at time $t \geq t^* + M$,

$$\begin{aligned}\bar{Y}_t^0 - \bar{Y}_t^1 &= \frac{1}{\sum_i (1 - S_i)} \sum_{i=1}^n Y_{i,t} \cdot (1 - S_i) - \frac{1}{\sum_i S_i} \sum_{i=1}^n Y_{i,t} \cdot S_i \\ &= \mathbb{E}[Y_t \mid S = 0] - \mathbb{E}[Y_t \mid S = 1] \\ &\quad + \frac{1}{n} \sum_{i=1}^n \left(\frac{\sum_{i=1}^n Y_{i,t} \cdot (1 - S_i)}{\sum_i (1 - S_i)} - \frac{\sum_{i=1}^n Y_{i,t} \cdot S_i}{\sum_i S_i} - \left(\frac{\mathbb{E}[Y_t(1 - S)]}{1 - p} - \frac{\mathbb{E}[Y_t \cdot S]}{p} \right) \right)\end{aligned}$$

Which is natural because under Assumption 1, the difference in sample averages is equal to the difference in population averages plus sampling error. Under the research design, $\mathbb{E}[Y_t \mid S = s] =$

$\mathbb{E}[Y_t \mid S = 0]$. So, when Mean Outcome Convergence holds,

$$\bar{Y}_t^0 - \bar{Y}_t^1 = \frac{1}{n} \sum_{i=1}^n \left(\frac{\sum_{i=1}^n Y_{i,t} \cdot (1 - S_i)}{\sum_i (1 - S_i)} - \frac{\sum_{i=1}^n Y_{i,t} \cdot S_i}{\sum_i S_i} - \left(\frac{\mathbb{E}[Y_t(1 - S)]}{1 - p} - \frac{\mathbb{E}[Y_t \cdot S]}{p} \right) \right)$$

Notice that when multiplied by $\sqrt{n}$, this term is exactly the pivotal quantity for the asymptotic distribution of $\widehat{\text{TATE}}(s, t)$, and the distribution is already derived in Proposition 5. Then,

$$\begin{aligned} \sqrt{n} (\bar{Y}_t^0 - \bar{Y}_t^1) &\xrightarrow{d} \mathcal{N}\left(0, \Sigma_t\right) \\ \bar{Y}_t^0 - \bar{Y}_t^1 &\xrightarrow{d} \mathcal{N}\left(0, \frac{1}{n} \Sigma_t\right) \end{aligned}$$

Where $\Sigma_t = \frac{p}{1-p} \mathbb{E}[Y_t \mid S = 0]^2 + \frac{1-p}{p} \mathbb{E}[Y_t \mid S = 1]^2 + \frac{\sigma_{0,t}^2}{1-p} + \frac{\sigma_{1,t}^2}{p}$, as in Proposition 5. To test Mean Outcome Convergence simultaneously for all post-switch time periods, I consider the joint distribution of these mean differences. Let

$$\hat{\boldsymbol{\theta}} := \begin{bmatrix} \bar{Y}_{t^*+M}^0 - \bar{Y}_{t^*+M}^1 \\ \bar{Y}_{t^*+M+1}^0 - \bar{Y}_{t^*+M+2}^1 \\ \vdots \\ \bar{Y}_{T-1}^0 - \bar{Y}_{T-1}^1 \\ \bar{Y}_T^0 - \bar{Y}_T^1 \end{bmatrix}'$$

Under Mean Outcome Convergence,

$$\hat{\boldsymbol{\theta}} \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{n} \cdot \boldsymbol{\Sigma}\right)$$

Where $\boldsymbol{\Sigma}$ is the variance matrix of mean differences over time, which is symmetric and square. The off-diagonal elements of $\boldsymbol{\Sigma}$ measure the within-group covariance in observed outcomes over time. When the research design holds, as $n \rightarrow \infty$, the Wald test statistic:

$$W = n \hat{\boldsymbol{\theta}}' \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\theta}} \sim \chi_q^2$$

where $q = T - (t^* + M) + 1$.

When M , the length of the carryover period, is unknown, it is more informative to construct confidence intervals around θ_t coefficients for all post-switch periods and examine when average outcomes across groups begin to converge (i.e. when $\theta_t \rightarrow 0$). Instead of comparing the single-valued Wald statistic to a critical value, we can construct a simultaneous confidence band as follows. First, construct $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\Sigma}}$ for all post-switch periods such that $q = T - t^*$. For a given significance level, α , solve for the set of $\hat{\boldsymbol{\theta}}$ values that would fail to reject a null hypothesis that $\hat{\boldsymbol{\theta}} = \vec{0}_q$. This is done by inverting the Wald statistic expression and setting it less than or equal to $\chi_{q, (1-\alpha)}^2$, and then solving for θ_t values. The resulting confidence bands have uniform coverage probability

$= 1 - \alpha$ across all $t \in q$.

4.3.1 Finite Sample Approach

In small samples, resampling methods such as bootstraps procedures can more efficiently generate valid confidence bands. I propose an adaptation of the multiplier bootstraps procedure of Callaway and Sant’Anna (2021). The algorithm is described below.

Multiplier Bootstraps Algorithm

When data are longitudinal, the multiplier bootstrap procedure of textcitecs2020 adapted for this setting is as follows: Let $\hat{\theta}_b^*$ be bootstrap draw $b \in B$ of $\hat{\theta}^*$. Also, let $\mathbf{Y}_i$ be a q length vector of outcomes for unit i . $\bar{\mathbf{Y}}^s$ is a q length vector of average outcomes for group s .

1. Compute $\hat{\theta}$ as the vector of sample mean differences between the two groups over time such that $\hat{\theta}$ is a $T - t^*$ length vector
2. Do the following B times:
 - (a) Draw mboot weights, $\{v^b\}_{i=1}^n$
 - (b) Compute $\hat{\theta}_b^* = \hat{\theta} + \frac{1}{n \cdot (1-\bar{p})} \sum_{i=1}^n v_i^b (\mathbf{Y}_i \cdot (1 - S_i) - \bar{\mathbf{Y}}^0) - \frac{1}{n \cdot \bar{p}} \sum_{i=1}^n v_i^b (\mathbf{Y}_i \cdot S_i - \bar{\mathbf{Y}}^1)$
 - (c) Form a bootstrap draw of the limiting distribution: $\mathbf{R}_b^* = \sqrt{n}(\hat{\theta}_b^* - \hat{\theta})$
3. Estimate the main diagonal of Σ , $\hat{\Sigma}^{boot}$ as follows. $\hat{\sigma}_t^{boot} = \frac{q_{0.75}(t) - q_{0.25}(t)}{1.349}$ where $q_p(t)$ is the p -quantile of R_t^* and the denominator is the inter-quartile range of the standard normal distribution ($z_{0.75} - z_{0.25}$)
4. Compute B test statistics as: $w_b^* = \|\mathbf{R}_b^* [\hat{\Sigma}^{boot}]^{-1}\|_\infty$
5. Construct $\hat{c}_{1-\alpha} = q_{1-\alpha}(w_b^*)$. The bootstrap critical value is the $1 - \alpha$ quantile of the B draws of w^* .
6. Construct the simultaneous confidence band: $\hat{C}_t = \left[\hat{\theta}_t \pm \hat{c}_{1-\alpha} \frac{\hat{\sigma}_t^{boot}}{n} \right]$

This method is computationally efficient because it re-weights pre-computed influence functions within each iteration of the bootstraps algorithm rather than recomputing the estimand. Additionally, this procedure constructs bootstrap standard errors using the interquartile range of the empirical distribution. Drawing from the 25th to 75th percentile of the distribution ensures that extreme bootstrap draws (which are more likely in small samples, or with small clusters) do not artificially inflate the width of the confidence bands. Compared to point-wise estimation of confidence intervals, the uniform confidence bands provide tighter estimates around the difference in means, increasing the probability of detecting non-zero differences in means across the two groups, and thus the credibility of the test.

5 Generalizing Results

5.1 Multiple Carryover Periods

So far, this paper has assumed that carryover effects of treatment can be constrained to one time period ($M = 1$). However, it might be empirically relevant to examine treatments whose effects carry over for $1 \leq M \leq T$ periods.

Proposition 7: *Under Assumptions 1-4, $TATE(s, t)$ is point identified by the expression in 4 for any M such that $0 \leq M \leq T$.*

Proof. Assume that the Stayers enter the model fully treated. This is synonymous to assuming that the Stayers are always-treated. Then, for the untreated cohort, for all t ,

$$\mathbb{E}[Y_t \mid S = 0] = \mathbb{E}[Y_t(\vec{1}_{M+1}) \mid S = 0].$$

For cohort s , treated at time t^* with M carryover periods, in all time periods j such that $j \leq t^* - M$, their untreated potential outcomes are observed. That is,

$$\mathbb{E}[Y_j \mid S = s, j \leq t^* - M] = \mathbb{E}[Y_j(\vec{0}_{M+1}) \mid S = s].$$

The target parameter for an arbitrary carryover period of length M is

$$\begin{aligned} TATE(s, j) &= \mathbb{E}[Y_j(\vec{1}_{M+1}) \mid S = s] - \mathbb{E}[Y_j(\vec{0}_{M+1}) \mid S = s] \\ &= \mathbb{E}[Y_j(\vec{1}_{M+1}) \mid S = 0] - \mathbb{E}[Y_j(\vec{0}_{M+1}) \mid S = s] \\ &= \mathbb{E}[Y_j \mid S = 0] - \mathbb{E}[Y_j \mid S = s] \end{aligned}$$

The first line is the definition of the target parameter, and the second line uses the result of proposition 2 to replace the average of fully-treated potential outcomes for group s with the average of fully-treated potential outcomes for group 0. As shown above, both terms in line 2 are observed. Line 3 replaces potential outcomes with observed outcomes. $\square$

Proposition 8: *Under Assumptions 1-4 and Assumption 5(a), $EMATE(s, t)$ is point identified by the expression in Equation 9 for any M such that $0 \leq M \leq T$.*

Proof. The proof is symmetric to the proof of proposition 7.

For an arbitrary carryover period of length $M \leq T$, for all $j \leq t^* - 1$,

$$\begin{aligned}
\text{EMATE}(s, j) &= \mathbb{E}[Y_j(\vec{0}_M, 1) \mid S = s] - \mathbb{E}[Y_j(\vec{0}_{M+1}) \mid S = s] \\
&= \mathbb{E}[Y_j(\vec{1}_{M+1}) \mid S = s] - \mathbb{E}[Y_j(\vec{0}_{M+1}) \mid S = s] + \mathbb{E}[Y_j(\vec{1}_{M+1}) - Y_j(\vec{0}_M, 1) \mid S = s] \\
&= \mathbb{E}[Y_j(\vec{1}_{M+1}) \mid S = s] - \mathbb{E}[Y_j(\vec{0}_{M+1}) \mid S = s] + \mathbb{E}[Y_{t^*}(\vec{1}_{M+1}) - Y_{t^*}(\vec{0}_M, 1) \mid S = s] \\
&= \mathbb{E}[Y_j(\vec{1}_{M+1}) \mid S = 0] - \mathbb{E}[Y_j(\vec{0}_{M+1}) \mid S = s] + \mathbb{E}[Y_{t^*}(\vec{1}_{M+1}) \mid S = 0] - \mathbb{E}[Y_{t^*}(\vec{0}_M, 1) \mid S = s] \\
&= \mathbb{E}[Y_j \mid S = 0] - \mathbb{E}[Y_j \mid S = s] + \mathbb{E}[Y_{t^*} \mid S = 0] - \mathbb{E}[Y_{t^*} \mid S = s]
\end{aligned}$$

The first line is the definition of the Within-Period treatment effect. The second line follows by adding and subtracting $\mathbb{E}[Y_j(\vec{1}_{M+1}) \mid S = s]$ and the third line applies Assumption 5(a) under which $\mathbb{E}[Y_{\tilde{t}}(\vec{1}_{M+1}) - Y_{\tilde{t}}(\vec{0}_M, 1) \mid S = s, t = \tilde{t}]$ is invariant in $\tilde{t}$. This means that we can replace this difference in time period j with the difference in time period t^* . The fourth equality holds by Proposition 1(a). Note that all terms in line 4 are observed, so the fifth line expresses these potential outcomes in terms of observed outcomes across cohorts. $\square$

Remark 5: It might be the case that the length of the carryover period is cohort specific. While estimators are derived here with potential outcomes indexed to M carryover periods for all groups, the Stayers' potential outcomes can be written with respect to $\tilde{M}$ such that $\tilde{M} \leq \infty$ carryover periods. The identification strategy is agnostic to the length of carryover for this group because units in the Stayer group enter the model fully treated. If the Stayers do not enter the model fully treated, but instead become first treated at $\tilde{t}$, then treatment effects are identified for all time periods j such that $\tilde{t} + \tilde{M} \leq j \leq t^* - 1$.

5.2 Average Treatment Effects

Under outcome convergence, and because all units are treated in the universal adoption setting, the treatment effects are identified not only for specific treatment cohorts, but also for the entire population. The Average Treatment Effect at time t is the unconditional expected difference in potential outcomes between groups, and is identified when $\text{TATE}(s, t)$ is the same across both groups. In other words, when the treatment effect at time t is the same for both the Switchers and the Stayers, and the two groups sample the entire population, then the Average Treatment Effect (ATE) is identified for the entire population.

Proposition 9: *Under Mean Outcome Convergence, when the Switchers and the Stayers are sampled from the entire population and $\mathbf{W}_i = \mathbf{\Omega}_i$,*

$$\begin{aligned}
\text{TATE}(t) &= \mathbb{E}[Y_t(1, 1) - Y_t(0, 0)] \\
\text{EMATE}(t) &= \mathbb{E}[Y_t(0, 1) - Y_t(0, 0)]
\end{aligned}$$

are identified as population-level average treatment effects, and equal to the cohort-specific treatment effects.

Proof. It is sufficient to show that for all $s \in \mathcal{S}$, $\text{TATE}(s, t)$ is invariant in s and $\text{EMATE}(s, t)$ is invariant in s .

By Proposition 1(a), with Mean Outcome Convergence for K periods,

$$\mathbb{E}[Y_t(1, 1) \mid S = s] = \mathbb{E}[Y_t(1, 1) \mid S = s'] \quad \text{for all } s, s' \in \mathcal{S}$$

When $\mathbf{W}_i = \boldsymbol{\Omega}_i$,

$$\begin{aligned} \mathbb{E}[Y_t(0, 0) \mid S = s] &= \mathbb{E}[\mathbf{W}_i \mid S = s]' \mathbf{F}_{\mathbf{t}, (0, 0)} \\ &= \mathbb{E}[\boldsymbol{\Omega}_i \mid S = s]' \mathbf{F}_{\mathbf{t}, (0, 0)} \\ &= \mathbb{E}[\boldsymbol{\Omega}_i \mid S = s']' \mathbf{F}_{\mathbf{t}, (0, 0)} = \mathbb{E}[Y_t(0, 0) \mid S = s'] \end{aligned}$$

where the first equality is the factor structure for untreated potential outcomes, the second equality holds because under proposition 9, $\mathbf{W}_i = \boldsymbol{\Omega}_i$. The third equality applies the latent variable balance result of Proposition 1 and maps the factor structure to untreated potential outcomes. This implies that when the latent factor loadings that drive treatment effect heterogeneity are the same as those that drive level differences in untreated outcomes, Mean Outcome Convergence implies that untreated potential outcomes are on average equal across groups. This is a strong result because for the Stayers, untreated outcomes are never observed.

In Section 3, the cohort-specific treatment effect is defined as:

$$\begin{aligned} \text{TATE}(1, t) &= \mathbb{E}[Y_t(1, 1) - Y_t(0, 0) \mid S = 1] \\ &= \mathbb{E}[Y_t(1, 1) - Y_t(0, 0) \mid S = 0] \\ &= \text{TATE}(0, t) \end{aligned}$$

The first equality is by definition of $\text{TATE}(s, t)$, and the second uses the latent variable balance result of Proposition 1(a). The third equality is again by definition of $\text{TATE}(s, t)$. This equality demonstrates that the cohort-specific treatment effect is invariant to cohort membership for cohorts 0 and 1 (Stayers and Switchers). When these cohorts span the universal set, cohort-specific treatment effects are invariant to cohort membership for all groups such that

For any $s, s' \in \mathcal{S}$,

$$\text{TATE}(s, t) = \text{TATE}(s', t)$$

Thus, $\text{TATE}(s, t)$ with respect to any $s \in \mathcal{S}$ is equal to the population-level average treatment effect, $\text{TATE}(t)$.

By symmetry, the same result holds for the Extensive Margin Treatment Effect (EMATE). $\square$

By Proposition 9, in universal adoption settings where the Switchers and Stayers are sampled from an entire population, it is possible to identify the Average Treatment Effects, which is a strong result. Inference under Proposition 9 is only possible when the universe of observations is

sampling is agnostic to treatment status, and when Universal Adoption actually applies to the entire population.

In practice, the Switchers and Stayers might be a subset of the sample that is treated, through targeted policies, selection into treatment, or other factors. In this case, even though treatment effects are identified for the two cohorts, and invariant in cohort membership within the treated, they are not necessarily equal to the population-level ATE. In these cases, observations are sampled conditional on exposure to treatment, and violates the premise of Proposition 9. Many empirically relevant policies do satisfy the requirements of Proposition 9: Federally enacted policies, regional pollution exposure, and local economic shocks are all examples of policies that expose entire populations to treatment, and candidate settings for with Proposition 9 can be applied.

5.3 Alternative Approaches

The estimators in Section 4 can alternatively be expressed as influence function-based estimators. The benefit of this approach is that the functional form, asymptotic properties, and weighting schemes for this function are known, and alternative approaches to achieving latent variable balance can be intuitively applied. If, for example, mean outcome convergence can only hold conditional on some covariates, X , or after re-weighting according to propensity scores, the implementation of propensity score weighting, doubly robust estimators, and regression adjustments can be derived from the influence function. The influence function for the Total Average Treatment Effect is derived here.

For $t \leq t^* - 1$, the population-level total treatment effect is:

$$\begin{aligned} \text{TATE}(s, t) &= \mathbb{E}[Y_t \mid S = 0] - \mathbb{E}[Y_t \mid S = s] \\ &= \mathbb{E}\left[\frac{1}{Pr(S=0)} \cdot Y_t \cdot (1 - S) - \frac{1}{Pr(S=1)} Y_t \cdot S\right] \\ \text{Let } p &= Pr(S = s). \\ &= \mathbb{E}\left[\frac{1 - S}{1 - p} Y_t - \frac{S}{p} Y_t\right]. \end{aligned}$$

Taking the sample analog,

$$\widehat{\text{TATE}}(s, t) = \frac{1}{n} \sum_{i=1}^n \left(\frac{1 - S_i}{1 - \hat{p}} - \frac{S_i}{\hat{p}} \right) Y_{i,t} = \frac{1}{n} \sum_{i=1}^n \left(\frac{\hat{p} - S_i}{\hat{p}(1 - \hat{p})} \right) Y_{i,t} \quad (24)$$

Where $\hat{p} = \bar{S} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{S_i = s\}$.

The natural two-step procedure is as follows:

1. Compute $\hat{p} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{S_i = 1\}$, which is the proportion of the sample in the switchers group.
2. Plug in the estimate from (1) to get the treatment effect, which is the average over the entire sample of $\frac{\hat{p} - S_i}{\hat{p}(1 - \hat{p})} Y_{i,t}$.

The influence function approach extends beyond adding covariates as from equation 24, it is clear that alternative causal inference approaches such as Propensity Score Weighting, Doubly Robust Estimators, and Machine Learning, can be applied to this estimand.

5.4 Adding Covariates

In some settings, mean outcome convergence might only be satisfied conditional on some observable covariates, X . Consider a simple case where X is a discrete and binary characteristic such as gender. If mean outcomes differ across genders, and the two groups have different proportions of men and women, then in general, Mean Outcome Convergence would not be satisfied. However, conditional on $X \in \{x_1, x_2\}$,

$$\mathbb{E}[Y_t((\vec{1}_{t^*+M:T}) | X = x_1, S = s)] = \mathbb{E}[Y_t((\vec{1}_{t^*+M:T}) | X = x_1, S = 0)]$$

and,

$$\mathbb{E}[Y_t((\vec{1}_{t^*+M:T}) | X = x_2, S = s)] = \mathbb{E}[Y_t((\vec{1}_{t^*+M:T}) | X = x_2, S = 0)]$$

Then Conditional Mean Outcome Convergence is satisfied: for all $t \geq t^* + M$,

$$\mathbb{E}[Y_t((\vec{1}_{t^*+M:T}) | X)] \perp S \quad (25)$$

As before, let p be the unconditional probability of a unit being in group s such that $p = Pr(S = 1)$. Additionally, let $p(x)$ be the conditional probability of a unit being in group s such that $p(x_j) = Pr(S = s | X = x_j)$. Notice that

$$\begin{aligned} \mathbb{E}[Y_t((\vec{1}_{t^*+M:T}) | S = s)] &= \mathbb{E}[\mathbb{E}[Y_t((\vec{1}_{t^*+M:T}) | X = x, S = s) | S = s]] \\ &= \mathbb{E}\left[\mathbb{E}[Y_t((\vec{1}_{t^*+M:T}) | X = x_1, S = s)] + \mathbb{E}[Y_t((\vec{1}_{t^*+M:T}) | X = x_2, S = s)] \middle| S = s\right] \\ &= \sum_{j \in \{1,2\}} Pr(S = s | X = x_j) \frac{Pr(X = x_j)}{Pr(S = s)} \mathbb{E}[Y_t((\vec{1}_{t^*+M:T}) | X = x_j, S = s)] \\ &= \sum_{j \in \{1,2\}} Pr(X = x_j) \frac{p(x_j)}{p} \mathbb{E}[Y_t((\vec{1}_{t^*+M:T}) | X = x_j, S = s)] \\ &= \sum_{j \in \{1,2\}} Pr(X = x_j) \frac{p(x_j)}{p} \mathbb{E}[Y_t(\vec{1}_{t^*+M:T}) | X = x_j] \\ &= \sum_{j \in \{1,2\}} \mathbb{E}\left[\frac{p(x_j)}{p} Y_t(\vec{1}_{t^*+M:T})\right] \end{aligned} \quad (26)$$

The first line holds by the Law of Iterated Expectations. The second line holds because when X is discrete, the conditional expectation over X is equal to the sum of the conditional expectations over the set of values x_j multiplied by the probability of x_j . The third line expands this. The

fourth line holds by Bayes' rule, and the fifth holds because conditional on X , $E[Y_t(\cdot)]$ is mean independent of S under the expression in 25. The expression in the sixth line is obtained by noting that $\sum_j Pr(X = x_j) \mathbb{E}[Y | X = x_j] = E[Y]$. Symmetrically,

$$\begin{aligned} \mathbb{E}[Y_t(\vec{1}_{t^*+M:T}) | S = 0] &= \mathbb{E}[\mathbb{E}[Y_t(\vec{1}_{t^*+M:T}) | X = x, S = 0] | S = 0] \\ &= \mathbb{E}\left[\frac{(1-p(x_1))}{1-p} Y_t(\vec{1}_{t^*+M:T})\right] + \mathbb{E}\left[\frac{(1-p(x_2))}{1-p} Y_t(\vec{1}_{t^*+M:T})\right] \end{aligned}$$

Then the expression in the last line of 26 can be written:

$$\sum_{j \in \{1,2\}} \mathbb{E}\left[\frac{p(x_j)}{p} Y_t(\vec{1}_{t^*+M:T})\right] = \sum_{j \in \{1,2\}} \mathbb{E}\left[\frac{p(x_j)}{p} \cdot \frac{1-p}{1-p(x_j)} Y_t(\vec{1}_{t^*+M:T}) \middle| S = 0\right]$$

When Conditional Mean Outcome Convergence according to 25 is satisfied, for all $t \geq t^* + M$,

$$\mathbb{E}[Y_t(\vec{1}_{t^*+M:T}) | S = s] = \mathbb{E}\left[\frac{1-p}{p} \sum_{j \in \{1,2\}} \frac{p(x_j)}{1-p(x_j)} Y_t(\vec{1}_{t^*+M:T}) \middle| S = 0\right] \quad (27)$$

The estimated total treatment effect is identified for all pre-switch periods, $t \leq t^* - 1$ as

$$\begin{aligned} \text{TATE}(s, t) &= \mathbb{E}[Y_t(\vec{1}_{t^*+M:T}) | S = s] - \mathbb{E}[Y_t(\vec{0}_{t^*+M:T}) | S = s] \\ &= \mathbb{E}\left[\frac{1-p}{p} \sum_{j \in \{1,2\}} \frac{p(x_j)}{1-p(x_j)} Y_t(\vec{1}_{t^*+M:T}) \middle| S = 0\right] - \mathbb{E}[Y_t(\vec{0}_{t^*+M:T}) | S = s] \\ &= \frac{1-p}{p} \mathbb{E}\left[\sum_{j \in \{1,2\}} \frac{p(x_j)}{1-p(x_j)} Y_t \middle| S = 0\right] - \mathbb{E}[Y_t | S = s] \end{aligned} \quad (28)$$

The first line is the definition of total treatment effect, and equality with the second line holds by substituting the equation in 27. Because $Y_t(\vec{1}_{t^*+M:T})$ is observed for the Stayers in all time periods, and $Y_t(\vec{0}_{t^*+M:T})$ is observed for the Switchers in all pre-switch time periods, the last line maps these potential outcomes to observed outcomes.

Remark 6: Conditional Mean Outcome Convergence designs require an additional overlap condition: $0 \leq p(x_j) \leq 1$ for all $x_j \in X$. This condition is natural in panel data models that include covariates, but is taken as given in the derivation above.

Because X is discrete and binary, a simple covariate balancing technique identifies the treatment effect. When X is continuous, various covariate balancing methods such as Regression Adjustment, and Inverse Propensity Score Weighting (IPW) can be applied. In Appendix B, I derive the

IPW estimator for $TATE(s, t)$ when Mean Outcome Convergence holds conditional on X . Modern techniques such as the Doubly Robust estimator and machine learning tools are more efficient when balancing continuous covariates. These methods are not discussed in this paper but can be implemented for Conditional Mean Outcome Convergence designs.

6 Application

While the Mean Outcome Convergence research design of this paper limits the applicability of the approach in empirical settings, as contended throughout the paper, contingent on this, the design is highly flexible to empirical settings. To demonstrate the methods of this paper, I examine the role of the Pregnancy Discrimination Act of 1978 on women’s employment in the United States.

6.1 Background

The Pregnancy Discrimination Act of 1978 (PDA) is a federal policy that modified the Civil Rights Act of 1964, prohibiting workplace discrimination on the basis of pregnancy or pregnancy-related conditions such as childbirth and postpartum conditions. It requires employers to extend to pregnant women the same workplace protections as disabled workers including health insurance coverage for pregnancy, reasonable workplace accommodations, and protection from discriminatory hiring and firing practices. However, because pregnancy is a transitory condition with unknown probability of onset, employers consider the costs of these accommodations at the point of hiring. In practice, enactment of the PDA increased the expected costs that employers faced when hiring childbearing-aged women relative to hiring men of the same age group. Because discrimination on the hiring margin is much more implicit than on the job displacement margin, PDA protections counter-productively decreased the probability of hiring women.

While PDA was enacted and enforced at the federal level, many state-level policies providing the same, or greater, protections against discrimination were enacted prior to its passage. Variation in the timing of enactment of these state laws has been exploited in studying the effect of workplace discrimination laws on women’s labor market outcomes (Gruber 1994; Timpe 2024). Consistent with staggered-adoption settings for Difference-in-Differences literature, treatment effects are identified for states treated prior to 1978, but not for states directly affected by the federal policy. However, these late adopter states often represent a high-impact subgroup that is meaningful to study. Di Giovan Paolo and Marcolin (2025) use the Difference-in-Differences-in-Reverse strategy to examine the impact of PDA on women’s employment by comparing the outcomes of states affected by PDA (Switchers) to those with pre-existing policies in 1978 (Stayers). In this application, I demonstrate that when the share of women employed converges across Switcher and Stayer states, causal inference does not rely on a parallel trends assumption. Furthermore, the effect of PDA on labor market outcomes is likely changing over time for both Switchers and Stayers. The Mean Outcome Convergence approach validates the causal interpretation of coefficients in this setting while DDR estimators do not.

Treatment effects are dynamic in both calendar time and event time in this application. Firms' responses to the new legal environment might undergo a transitory learning period before they reach equilibrium in their discriminatory hiring practices. This learning period is directly related to the amount of time that the state has enforced PDA-related policies. In other words, treatment effects vary with time relative to first treatment (event time). In addition to this, treatment is affected by exogenous, time-varying conditions regardless of the length of time from first treatment. The extent to which PDA discourages firms from hiring women is not agnostic to cultural attitudes, political environment, economic trends, and aggregate productivity. These factors vary with time, and possibly affect all states, not just Switchers. Thus, it is relevant to consider causal inference when the treatment effect is changing over calendar time for all treated groups.

When evaluating the additional costs associated with hiring childbearing-age women, firms consider the both the laws and the strength with which these laws are enforced within the state. If the strength of enforcement itself is changing in early-switch periods, then the Switcher states are experiencing a different level of treatment in these time periods. For example, if legal agencies in Switcher states exercise leniency when PDA is first enacted, and increase penalties for violations as the law becomes more commonplace over time, then PDA protections in the first treated periods are *less* in magnitude than PDA protections in later periods. See Remark 4 for a discussion of this in theoretical terms. Because of this possibility, I caution that the estimated treatment effects are a lower bound on the magnitude of treatment effects. If treatment effects are negative, as is expected in this setting, then in absolute terms, the estimates provide an upper-bound on the true effect.

6.2 Data

Analysis of the policy is conducted using the Current Population Survey (CPS) Annual Supplement of Economic Conditions (ASEC), which is conducted in March of every year and includes detailed labor market information including earnings, hours of employment, weeks of employment/unemployment, and industry (Flood et al. 2025). ASEC surveys also includes questions about household composition and ages of all children which can be used to construct a measure of fertility.

States are sorted into Switcher or Stayer groups based on treatment timing. States that adopted policies with protections similar to PDA prior to 1978 are grouped as Stayers. States with no existing policies establishing pregnancy as a protected class in the workplace are grouped as Switchers. Di Giovan Paolo and Marcolin (2025) select as Stayers the states that passed a policy prior to 1973, and further restricts Stayers to the subset of these states that are identified in ASEC survey data during the time period. Replicating their grouping, I define the following states as Stayers in my data: Connecticut, Massachusetts, New Jersey, and Pennsylvania. Di Giovan Paolo and Marcolin (2025) define as Switchers the states that did not have a PDA-like policy prior to 1978, and further restrict the sample to states identified in ASEC survey data. The Switcher states are: Alabama, Arkansas, Florida, Georgia, Indiana, Kentucky, Louisiana, Mississippi, North Carolina,

Ohio, Oklahoma, South Carolina, Tennessee, and Texas.

For Switcher and Stayer states, I include in the analysis women between the ages of 18 and 35, the age group considered to be of child-bearing age during the time period. Gruber (1994) conducts analysis based on women of ages 20-40. Expanding the sample to ages 18-40 does not qualitatively change results. All results are weighted using the ASEC individual-level weights, once again following the procedures of Di Giovan Paolo and Marcolin (2025). Table 1 contains survey-weighted summary statistics for switcher and stayer states for the years used in the analysis.

Table 1: Summary Statistics for Women Aged 18-35

	Pre			Post		
	Mean	SD	Obs	Mean	SD	Obs
Panel A: Switchers						
Age	25.645	5.132	56962	26.358	5.110	79458
Married	0.656	0.475	56962	0.555	0.497	79458
Pregnant	0.101	0.302	56962	0.087	0.282	79458
Employed	0.673	0.469	56962	0.749	0.434	79458
Laborforce	0.560	0.496	56962	0.681	0.466	79440
Earnings	3667.701	3171.249	36834	8594.107	7624.538	57381
Panel B: Stayers						
Age	25.832	5.179	19937	26.399	5.157	29679
Married	0.580	0.494	19937	0.479	0.500	29679
Pregnant	0.089	0.285	19937	0.077	0.266	29679
Employed	0.635	0.481	19937	0.755	0.430	29679
Laborforce	0.542	0.498	19937	0.682	0.466	29674
Earnings	4281.991	3619.329	12290	9893.947	8766.012	21845

¹ Data reported for Switcher states: Alabama, Arkansas, Florida, Georgia, Indiana, Kentucky, Louisiana, Mississippi, North Carolina, Ohio, Oklahoma, South Carolina, Tennessee, and Texas and Stayer states: Connecticut, Massachusetts, New Jersey, and Pennsylvania, as defined by Di Giovan Paolo and Marcolin (2025).

² All summary statistics are computed using individual-level ASEC weights.

³ Marital status is determined by survey responses. Pregnancy is determined as having a child less than age 1 at the time of the survey. Employment is determined by at least one week of reported employment over the past year. Labor force status and annual earnings are self-reported.

The summary statistics table reports the survey-weighted mean, standard deviation, and number of observations for each variable of interest. The average age within the sample is similar across states. This is somewhat trivial since age is a component of the selection criteria for the sample. Women in Switcher states report being married at higher rates than those in Stayer states. They also have slightly higher rates of pregnancy. The pregnancy rate is computed by determining whether a woman has a child of age less than 1 at the time of the survey. This covers all births from April of the previous year to March of the survey year, and therefore measures pregnancies two years preceding the survey. This is a noisy measure of pregnancy, however it is the most informative as the ASEC fertility supplement is not available for this time period. Across pre and

post-switch periods, the difference in labor force participation rates remains steady (and not statistically significant) between Switchers and Stayers. The share of women employed is computed by determining whether a woman reported a non-zero number of weeks spent working in the past year. This is a better measure of employment than the CPS currently employed survey question because it measure labor market activity over the entire year, not just transient labor market activity in March, at the time of the ASEC survey. Notably, the percentage of women employed gainfully in the past 12 months is equal across groups in the post-switch period. Average annual earnings are higher for women in Stayer states and the gap is consistent between pre and post-switch periods. However, across both groups and both time periods, the standard deviation of this measure is very high, indicating significant variation in earnings income within groups.

The key outcome of interest is employment, which is measured by whether or not a woman aged 18-35 reported working for at least one week in the past year. The graphs below show raw trends in this outcome over the period of interest. The graph indicates that on average, the two groups are converging in key outcome of interest in the post-switch period. This will be confirmed in empirical testing in the next section. In Figure 1, I graph raw trends for a six-year window around the policy change.

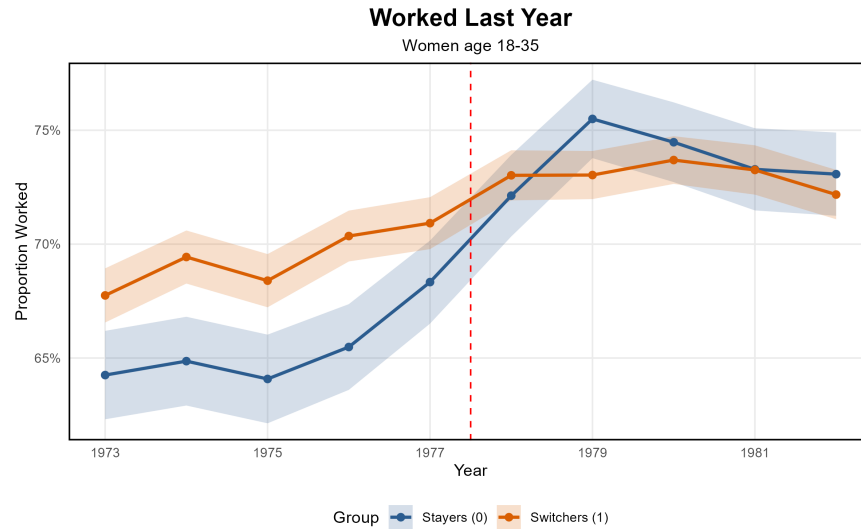


Figure 1: Key outcome: Share of women aged 18-35 who were employed. Outcome is measured by whether they worked for at least one week in the past year. PDA adoption for switchers occurs in 1978. Sample means and standard errors computed using ASEC individual survey weights. Shaded regions represent 95% confidence intervals using ASEC individual survey weights.

Extended Time Window

Recall that in the main identification result, the dimensions of heterogeneity that the model handles is equal to the number of post-switch periods for which Mean Outcome Convergence is observed. The model is equipped to identify treatment effects under greater degrees of heterogeneity when there are more post-switch periods for which Mean Outcome Convergence is observed. In Figure 2 below, I extend the post-switch time window to 1989 and the graph indicates that Mean

Outcome Convergence still holds.

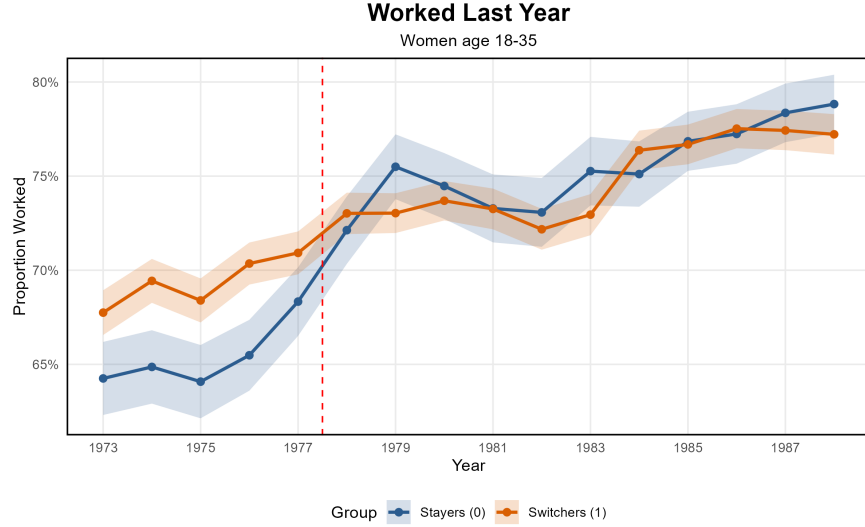


Figure 2: Key outcome: Share of women aged 18-35 who were employed. Outcome is measured by whether they worked for at least one week in the past year. PDA adoption for switchers occurs in 1978. Sample means and standard errors computed using ASEC individual survey weights. Shaded regions represent 95% confidence intervals using ASEC individual survey weights.

It is natural to ask: what if Mean Outcome Convergence holds for a short period of time, but average outcomes begin to diverge between groups over time. Whether the research design is violated in such cases is not immediately clear. On one hand, average outcomes diverging after some time might be evidence against latent variable balance in that only some combinations of G_t and $\mathbb{E}[\Omega_i | S]$ are equal across groups. On the other hand, the usual panel data causal inference issues with moving further away from the Switch time period apply. As units are observed at time periods further and further from the time frame of interest, other external factors such as related policy changes are more likely to be present, making the groups less comparable. As in the usual case, this does not always indicate that the groups are poor comparisons for each other, but that the groups observed at time periods far away from the switch are less likely to be on the same time trends due to external correlated changes. Researchers should exercise due diligence in examining outcomes for extended time windows, considering other policy changes that could be driving differences (or similarities) in outcomes.

Potential Mechanisms

In this section, I consider potential mechanisms that are driving the convergence in employment rates for women between Switcher and Stayer states.

In Figure 3, I attempt to investigate job displacement rates. The first figure of the panel graphs the share of women aged 18-35 who are active in the labor force. Panel (a) of Figure 3 shows that there are no significant differences in this measure across groups in the pre-switch period. In the post-switch period, labor force participation rates often overlap for Switcher and Stayer states. In

the second panel of this figure, I graph the share of women who are employed, conditional on being in the labor force. This figure is used to highlight job displacement as the outcome is inversely proportional to the unemployment rate. Panel (b) of Figure 3 shows that after the enactment of PDA, women in the labor force in Switcher states are employed at lower rates compared to those in Stayer states. Taken together, these graphs indicate that job displacement is evident in Switcher states. The share of women in the labor force is the same across groups, and does not change discontinuously at the time of the switch. This indicates that women are looking for work at the same rate both before and after the switch. However, among those in the labor force, a greater share of women are not employed in Switcher states in the post-switch period. One interpretation of this difference is that women in Switcher states are experiencing job displacement at the time of the switch, while PDA-displaced employees in Stayer states have already exited the workforce. It is important to note that because panel (b) is a conditional proportion, the composition of women represented in the figure is changing over time (and changing at the rate of change of panel (a)), and should be interpreted with caution. A graph of the share employed over time is included in the supplementary appendix and shows that the unconditional share of women employed does not diverge as panel (b) of Figure 3, and closely tracks the main outcome in Figure 2.

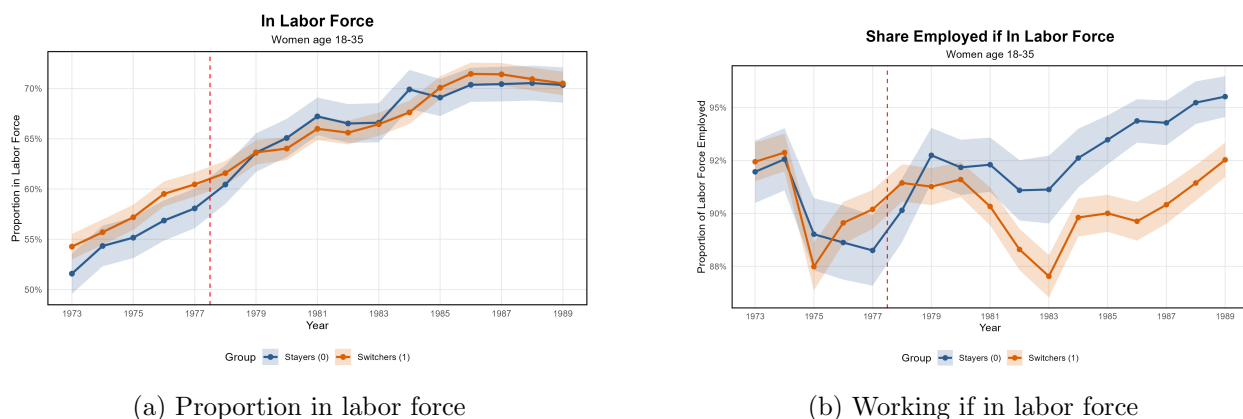


Figure 3: Both figures generated using self-reported labor force status. Figure (a) is labor force status for women age 18-35. Figure (b) is employment status among women in the labor force, aged 18-35. Shaded regions represent 95% confidence intervals using ASEC individual survey weights.

Figure 3 indicates that since labor force participation is smooth across the switch, the changes in employment are not likely to be driven by women opting to leave the labor force. I confirm this with Figure 4, which graphs the share of women pregnant over time. The measure is constructed based on whether a woman reports having a child that is less than one year old at the time of the survey (March of each year). It measures pregnancies that resulted in births between April of the previous year to March of the current year. This is a noisy measure of pregnancy because it does not consider pregnancies over the past year that did not result in a live birth. More accurate measures of fertility, in the CPS ASEC Fertility Supplement, are not available from CPS data for this time period. Figure 4 demonstrates that the measured fertility rate is smooth between pre- and post-switch periods, and is further trending similarly between Switcher and Stayer states. This

graph supports the idea that enactment of PDA decreased women's employment by discouraging firms from hiring childbearing-aged women, rather than by encouraging women in this age group to bear more children. In the appendix, I also include a plot for annual earnings for those who are employed. As with the figures in this section, the plot shows no discontinuous changes around the switching time period.

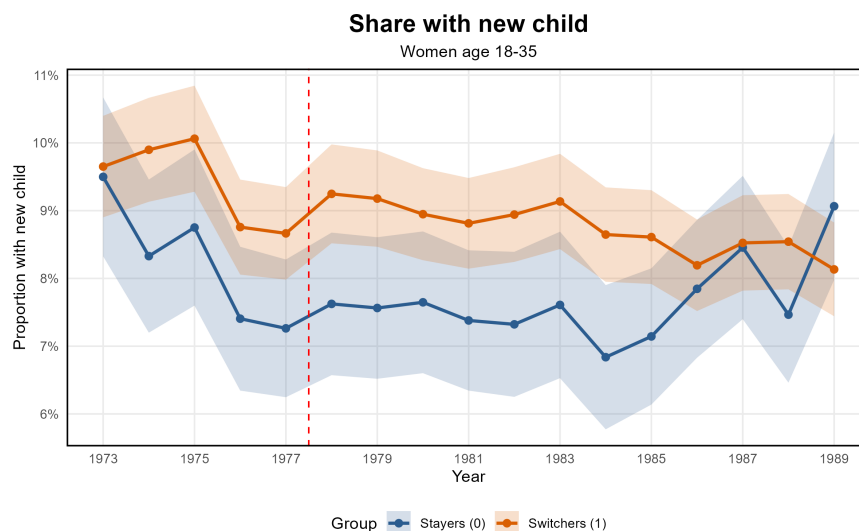


Figure 4: Proportion of women age 18-35 with youngest child less than 1 in the next survey year. Covers births from April of previous year to March of current year. Sample means and standard errors computed using ASEC individual survey weights. Shaded regions represent 95% confidence intervals using ASEC individual survey weights.

6.3 Research Design

While Figures 1 and 2 do not indicate differences in average outcomes between Switcher and Stayer states in the post-switch period, a formal test of the research design is necessary. The raw trends do not provide sufficient information about the population averages across groups, which is the condition for Mean Outcome Convergence to hold. Additionally, the length of the carryover period is unknown when examining raw trends alone.

To test the research design, I employ the multiplier bootstraps procedure detailed in Section 4.3. It is important to note that since CPS data are repeated cross-sections, the algorithm's weights for each unit i only affect outcomes at the time period in which unit i is observed. This means that the procedure does not account for individual unobservables, Ω_i , but rather the average at the group level, $\mathbb{E}[\Omega_i \mid S]$. Confidence bands around the difference in mean employment shares across groups, constructed using the multiplier bootstraps algorithm, are presented in Figure 5.

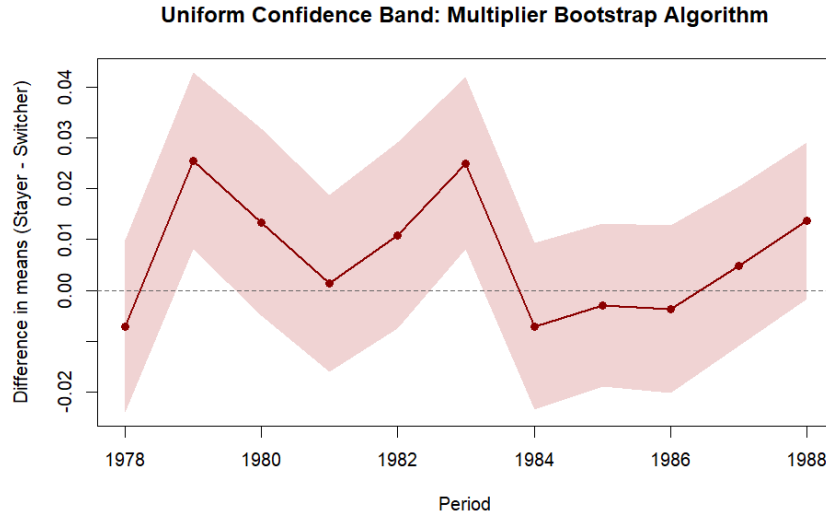


Figure 5: Difference in mean outcome between Stayers and Switchers in the post-switch period. Shaded regions represent 95% uniform confidence bands constructed using the multiplier bootstraps algorithm. All measures are computed using ASEC individual survey weights.

Figure 5 graphs uniform confidence bands around the difference in means between Stayers and Switchers. Under the research design, the mean difference should be statistically indistinguishable from zero after M carryover periods. The graph indicates that mean differences are significant in two post-switch periods, 1979 and 1983. After 1983, the estimated mean difference is close to zero with a tight confidence interval, meaning that after 1983, the algorithm fails to detect any significant differences in outcomes between the two groups at the 5% significance level. Figure 5 supports the research design and indicates that Mean Outcome Convergence holds for years including and after 1984. Since the switch occurs in 1978, this indicates a carryover period of 5 years ($M = 1983 - 1978 = 5$).

6.4 Estimation and Results

Estimation of Total Treatment Effects (TATE(s,t)) is computed from the expression in ?? as

$$\widehat{\text{TATE}}(1, t) = \frac{1}{\sum_i (1 - S_i)} \sum_{i=1}^n Y_{i,t} \cdot (1 - S_i) - \frac{1}{\sum_i S_i} \sum_{i=1}^n Y_{i,t} \cdot S_i$$

Where $S_i = 1$ if an observation belongs to a Switcher state and 0 otherwise. $Y_{i,t}$ for each unit i is an indicator for whether they were employed for more than one week in year t . Standard errors for these coefficients are computed using the formula in 23. Estimated coefficients and confidence intervals are shown in Figure 6.

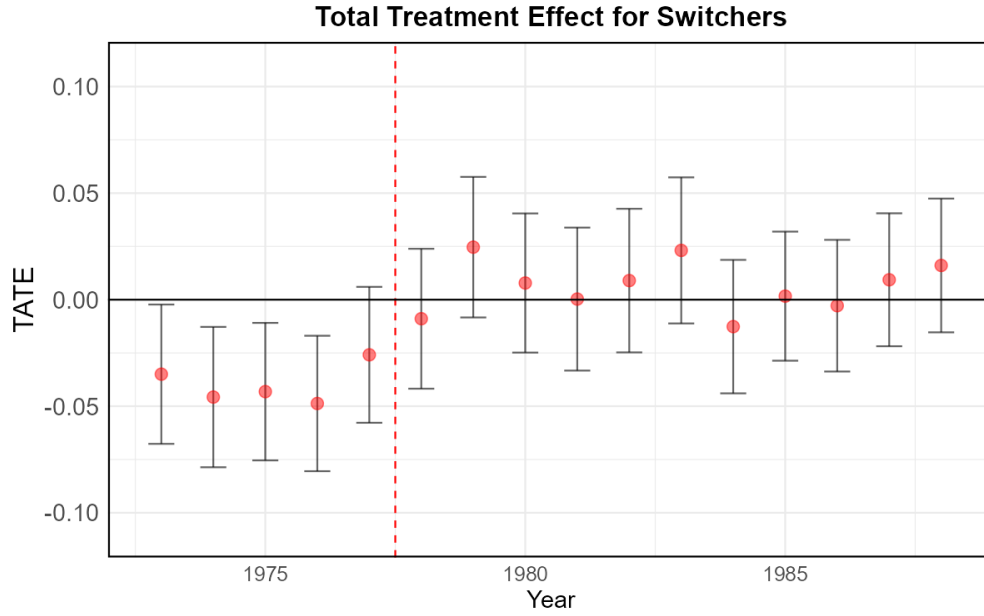


Figure 6: TATE(s,t) coefficients for Switchers at time t . Bars around each point estimate represent 95% confidence intervals with standard errors computed based on the estimation method in Section 4. Confidence intervals are computed adjusting for sampling error from $\hat{p}$. All point estimates and standard errors are computed using ASEC individual survey weights.

Figure 6 shows the estimated treatment effects for Switchers in the pre-switch periods, and the differences in treated outcomes across groups in the post-switch periods. In the pre-switch periods, the TATE(s,t) coefficients are negative and significant. This indicates that if Switcher states had enacted PDA-like protections prior to 1978, women’s employment in these states would have fallen by an average of 4.15 percentage points. This point estimate is comparable to the main specification result of Di Giovan Paolo and Marcolin (2025), which reports a 4.6 percentage point reduction in employment in the Switcher states.

The point-wise estimates for each pre-switch time period are interpretable as the effect of the policy on the switchers, if it were adopted in that time period. When assuming $M = 5$ carryover periods, the largest treatment effect, in 1970, is interpreted as follows. If the Pregnancy

Discrimination Act was enacted in 1965, Florida would reach their fully-treated status by 1970. Therefore, PDA enactment in 1965 would lead to women aged 18-35 would experiencing a decrease in employment of 6.56 percentage points, on average in 1970. The separate coefficients for each time period lend to rich welfare analysis. For example, in Florida in 1970, 45.94% of women aged 18-35 in the sample were employed. If PDA were enacted in Florida in 1965 instead of in 1978, a quick back-of-the envelope calculation indicates:

$$\begin{aligned}\widehat{Y}_{1970,FL}^{PDA} &= Y_{1970,FL} + \widehat{TATE}(1, 1970) \\ &= 45.94 - 6.56 = 39.38\end{aligned}$$

The decreases in women's employment driven by the Pregnancy Discrimination Act would have resulted in an employment rate of 39.38% for women aged 18-35 in Florida, if it were enacted in 1970. This is an approximately 14.28% decrease in women's employment. These estimates can inform state policymakers in designing policies that alleviate the adverse effects of PDA, to establish willingness-to-pay thresholds, or state-subsidized programs that mitigate the costs to employers.

As an exercise, I plot total treatment effects with confidence intervals constructed without adjusting for group sampling error, which assumes that the proportion of treated units is known and accurate in the sample such that $\hat{p} = p$. This figure is included in the appendix (Figure C.3). As one would expect, the confidence intervals constructed using this method are narrower than those in Figure 6. The discussion below examines why, even when group sampling error is small, the adjusted confidence intervals in Figure 6 are preferable.

Notice that the plotted estimates in the post-switch period in Figure 6 are close to 0. This is expected under Mean Outcome Convergence; the difference in mean outcomes in the post-switch period should be close to zero. However, notice that the point-wise confidence intervals in Figure 6 are significantly wider than those in Figure 5. The confidence intervals in Figure 6 indicate that the differences in outcomes are not statistically significant for any time periods, while the multiplier bootstraps procedure detects differences in 1979 and 1983. This highlights the low power of point-wise hypothesis testing in the research design. The distinction is clear when considering the hypotheses that each approach is testing. The confidence intervals in Figure 6 are constructed to maximize the power of a test detecting whether treatment effects are significant. The test is optimal for:

$$\begin{aligned}H_0 &: TATE(s, t) = 0 \\ H_\alpha &: TATE(s, t) \neq 0\end{aligned}$$

When the (wider) confidence intervals do not contain 0, this provides strong evidence against the null hypothesis that the differences in means between the two groups is equal to 0. This is appropriate when pursuing strong evidence that treatment effects are significant. However, note that in testing the research design, we are in pursuit of strong evidence that the difference in means is equal to 0. In this case, a wide confidence band that contains zero leads to under-detection of

significant differences in means. Instead, the appropriate test constructs narrow confidence bands around the difference in means and pursues evidence that this band *does* contain 0. This is why the uniform confidence band is better equipped to test the research design compared to point-wise tests. Another method of increasing the power of the research design test would be to use Equivalence testing methods, which construct small intervals around 0 and use one-sided tests to determine whether the difference in means is within this interval. See Bilinski and Hatfield (2026) and Roth (2022) for detailed discussions on Equivalence Testing in econometric models.

6.5 Discussion

The Pregnancy Discrimination Act, while enacted to protect pregnant women in the workplace, increased the firm’s expected cost of hiring childbearing-age women, and thus in practice discouraged firms from hiring members of this demographic group. The extent of discriminatory practices driven by the act is driven by many time-varying characteristics such as economic conditions. During periods of economic growth, it is likely that the productivity gains from hiring more workers exceed the expected costs of hiring childbearing-age women, attenuating the negative effect of the law on women’s employment. On the other hand, during economic recessions, the effect of the act would likely be exacerbated. These time-varying conditions drive the treatment effect for all states, including Stayer states that are used as comparison units. While traditional common trends assumptions require researchers to assume away these effects in DDR models, my approach exploits Mean Outcome Convergence between Switchers and Stayers and identifies the effect of the act when this time heterogeneity is present. The event study coefficients indicate that the treatment effect is negative and significant, and comparable in magnitude to other methods (Di Giovan Paolo and Marcolin 2025). When testing the research design, the multiplier bootstrap algorithm proposed in this paper is higher powered in detecting true differences in mean outcomes across groups when compared to the point-wise approach.

7 Conclusion

In this paper, I examine a universal adoption setting for a panel data model in which an untreated control group doesn’t exist. I study comparisons between two groups, the Switchers and the Stayers, when they feature Mean Outcome Convergence in the post-switch period, meaning that their average outcomes are statistically indistinguishable. Under Mean Outcome Convergence, identification relies on relatively mild assumptions and allows for complex factor structures for potential outcomes.

When Mean Outcome Convergence holds eventually, potential outcomes exhibit limited carry-over, meaning that treatment affects future potential outcomes for a limited amount of time. The limited carryover assumption is a relaxation of the more general “no carryover” assumption often invoked in causal inference literature. In unequal baseline settings such as the universal adoption setting, no carryover assumptions limit researchers to settings with non-cumulative treatment ef-

fects. By shifting to a limited carryover assumption, I propose an unequal baseline model that is robust to dynamic treatment effects. The proposed estimators are valid even when Switchers and Stayers have different carryover periods. Additionally, the structure of potential outcomes allows for heterogeneity in treatment effects and in untreated potential outcomes, and further allows these dimensions to be distinct from each other. Because the validity of comparisons between groups is discerned based on treated outcomes, treatment effects are identified even when there is no overlap between the drivers of treatment effect heterogeneity and the drivers of untreated potential outcomes.

I begin by demonstrating that under a Latent Unconfoundedness Assumption, Mean Outcome Convergence implies mean equality in latent variables across groups in a factor model of potential outcomes. Given this result, I demonstrate that the counterfactual treated potential outcomes for the Switchers in the pre-switch period are identified by the observed treated outcomes for the Stayers in this time period. Treatment effects can then be identified using simple estimators. Because my approach allows treatment effects to cumulatively affect potential outcomes, it motivates identification of two distinct treatment effects: the total treatment effect and the extensive margin treatment effect (effect of first treatment). Researchers should distinguish between these effects in unequal baseline models as their interpretations and implications for policy analysis and other inference differ from each other. For example, when determining the cost effectiveness for treatment, it is common to compare the efficacy of a treatment to its cost. If an extensive margin effect is estimated, the true efficacy of the treatment would be understated. The methods in this paper distinguish between total and extensive margin treatment effects and propose estimators for each; this is key to accurate analysis and interpretation of treatment effects.

The flexibility of the model relies on Mean Outcome Convergence in the data. Therefore, it is important that a test of the research design is powered to detect differences between the two groups when they exist. In Section 4.3, I propose an adaptation of the multiplier bootstraps algorithm of Callaway and Sant’Anna (2021) that produces uniform confidence bands to simultaneously test Mean Outcome Convergence across the post-switch period. This procedure produces tight confidence bands around the estimand, and is better equipped to detect statistical differences when they exist, compared to point-wise confidence intervals. I demonstrate this in the application, where point-wise intervals are not able to detect mean differences in post-switch time periods where they exist.

I demonstrate the proposed methods by examining the effect of the Pregnancy Discrimination Act (PDA) of 1978 on women’s employment by comparing states affected by the federal mandate to states that passed PDA-like policies prior to 1978. The estimates indicate that in Switcher states, PDA would have decreased women’s employment by an average of 4.15 percentage points if it were enacted within the 10-year pre-switch period. In the state of Florida, the largest point-estimate translates to a 14.28% reduction in women’s employment, a striking result but one that is comparable to other estimates in the literature. While conventional staggered treatment adoption designs can be used to study the effect of PDA and other federal policies on early adopters (i.e.

Stayers), they cannot provide insight to the effect on states that adopt these laws only when they are mandated at the federal level. I argue that these states are often representative of a high-impact group. These “late adopter” states often culturally or politically resist the law, lack infrastructure for adequate implementation, or differ in their legal environment compared to “early adopter” states. It is relevant both to economists and to policymakers to study treatment effects for these groups, and unequal baseline models are valuable in these settings.

While the approach of this paper is flexible to many sources of heterogeneity, it also presents a high price for that flexibility in that the Mean Outcome Convergence research design rules out many settings. When Mean Outcome Convergence can only be satisfied conditional on covariates, the Conditional Mean Outcome Convergence design should be used to balance covariates across groups. In some settings, the carryover period might be greater than the number of post-switch periods. While beyond the scope of this paper, more advanced computational techniques may be used to approximate a rate of convergence in outcomes, and identify treatment effects when outcomes eventually converge across groups. Relatedly, when covariates are available and closely related to unobserved variables, it is possible to derive moment conditions that reduce the dimensions of unobserved heterogeneity using these covariates. Other computationally advanced techniques can potentially lead to the inclusion of more settings that are ruled out in the simple Mean Outcome Convergence case. The methods derived in this paper establish a framework for causal inference in these models.

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Appendix

A Regression Approach

Other papers in the unequal baseline setting consider a non-parametric event study regression such as:

$$Y_{it} = \beta_1 S_i + \lambda_t + \sum_{j=1}^{t^*-1} \gamma_j^{pre} S_i \cdot \mathbf{1}(t = j) + \sum_{j=t^*+1}^T \gamma_j^{post} S_i \cdot \mathbf{1}(t = j) + \epsilon_{it} \quad (29)$$

Where Y_{it} are the observed outcomes for unit i at time t , S_i is an indicator that equals 1 for Switchers, and λ_t is a unit-invariant time coefficient. In an equal baseline panel data model (such as DiD), the γ^{post} might be interpreted as causal effects and γ^{pre} coefficients are used to assess the plausibility of the parallel trends assumption. Symmetrically, in the unequal baseline setting, the γ^{pre} coefficients are informative about treatment effects and γ^{post} coefficients are interpreted to support the validity of the design. It is common to normalize equal baseline models to $t^* - 1$ or t^* , which might imply that it makes sense to normalize the unequal baseline model to t^* or $t^* + 1$. However, under cumulative treatment effects, each of these normalizations yields different causal estimands.

Proposition A.1: *The estimated γ^{pre} coefficients from a non-parametric event study regression normalized to the switch period, t^* , as in Equation 29 are equal to the negative of the estimated $EMATE(s, t)$ values under Assumption 1-4 and Assumption 5(a).*

Proof. γ^{pre} coefficients for all $t \leq t^* - 1$ decompose to the difference in observed outcomes between group s and group 0, relative to their observed differences in the reference period, which is t^* :

$$\begin{aligned} \gamma_j^{pre} &= \mathbb{E}[Y_j - Y_{t^*} \mid S = s] - \mathbb{E}[Y_j - Y_{t^*} \mid S = 0] \\ &= -EMATE(s, j) \end{aligned}$$

□

Proposition A.2: *Under Assumptions 1-4, The estimated γ^{pre} coefficients from a non-parametric event study regression normalized to the first post-switch period, $t^* + 1$, identify different treatment effects at different $j \in [1, t^*]$.*

Proof. Consider an event study regression normalized to $t^* + 1$:

$$Y_{it} = \beta_1 S_i + \lambda_t + \sum_{j=1}^{t^*} \gamma_j^{pre} S_i \cdot \mathbf{1}(t = j) + \sum_{j=t^*+2}^T \gamma_j^{post} S_i \cdot \mathbf{1}(t = j) + \epsilon_{it} \quad (30)$$

The γ^{pre} coefficients for all $t \leq t^* - 1$ decompose to the difference in observed outcomes between group s and group 0, relative to their observed differences in the reference period, which is $t^* + 1$:

$$\gamma_j^{pre} = \mathbb{E}[Y_j - Y_{t^*+1} \mid S = s] - \mathbb{E}[Y_j - Y_{t^*+1} \mid S = 0]$$

Suppose $M = 1$. Under Mean Outcome Convergence with one carryover period, this decomposition becomes:

For all $j \leq t^* - 1$:

$$\begin{aligned} \gamma_j^{pre} &= \mathbb{E}[Y_j(0, 0) - Y_{t^*+1}(1, 1) \mid S = s] - \mathbb{E}[Y_j(1, 1) - Y_{t^*+1}(1, 1) \mid S = 0] \\ &= \mathbb{E}[Y_j(0, 0) \mid S = s] - \mathbb{E}[Y_j(1, 1) \mid S = 0] \\ &= -\text{TATE}(s, t) \end{aligned}$$

For $j = t^*$:

$$\begin{aligned} \gamma_j^{pre} &= \mathbb{E}[Y_{t^*}(0, 1) - Y_{t^*+1}(1, 1) \mid S = s] - \mathbb{E}[Y_{t^*}(1, 1) - Y_{t^*+1}(1, 1) \mid S = 0] \\ &= \mathbb{E}[Y_{t^*}(0, 1) \mid S = s] - \mathbb{E}[Y_{t^*}(1, 1) \mid S = 0] \end{aligned}$$

Which holds by mapping observed outcomes to potential outcomes and because under Mean Outcome Convergence, when $M = 1$, $\mathbb{E}[Y_{t^*+1}(1, 1) \mid S = s] = \mathbb{E}[Y_{t^*+1}(1, 1) \mid S = 0]$. The first three lines show that the γ_j coefficients for $j \leq t^* - 1$ are equal to the negative of total treatment effects, which has a different causal interpretation than the treatment effects recovered from the same coefficient under a different normalization (see Proposition A.1 where normalizing to t^* recovers $-\text{EMATE}(s, t)$). Furthermore, for time period t^* , the γ coefficient captures the (negative of) lingering effect of past treatment. When the event study regression is normalized to $t^* + 1$, the γ^{pre} coefficients recover different treatment effects for different time periods. It is not possible to infer the changes in treatment effects over time in this case because the coefficients at different points in time are not comparable to each other.

The problems compound when the event study coefficients are normalized to a time period before the Switchers are fully treated. To illustrate this, suppose Equation 30 is used to identify treatment effects in a model where $M > 1$. Since the normalization is with respect to $t^* + 1$, Switchers are treated for two periods at the reference time period, but are still experiencing event-

time dynamics. Decomposing the coefficient makes this clear.

For all $j \leq t^* - 1$:

$$\begin{aligned}
\gamma_j^{pre} &= \mathbb{E}[Y_j(\vec{0}_M, 0) - Y_{t^*+1}(\vec{0}_{M-1}, 1, 1) \mid S = s] - \mathbb{E}[Y_j(\vec{1}_M, 1) - Y_{t^*+1}(\vec{1}_M, 1) \mid S = 0] \\
&= \underbrace{\mathbb{E}[Y_{t^*+1}(\vec{0}_{M-1}, 0, 0) - Y_{t^*+1}(\vec{0}_{M-1}, 1, 1) \mid S = s]}_{\text{(negative) Effect of two periods of treatment}} \\
&\quad + \underbrace{\mathbb{E}[Y_j(\vec{0}_M, 0) - Y_{t^*+1}(\vec{0}_M, 0) \mid S = s]}_{\text{Time trend for untreated potential outcomes over } [j, t^*+1]} \\
&\quad - \underbrace{\mathbb{E}[Y_j(\vec{1}_M, 1) - Y_{t^*+1}(\vec{1}_M, 1) \mid S = 0]}_{\text{Time trend for fully treated potential outcomes over } [j, t^*+1]}
\end{aligned}$$

The first line maps the outcomes to potential outcomes and equality with the second line holds by adding and subtracting $\mathbb{E}[Y_{t^*+1}(\vec{0}_M, 0) \mid S = s]$. The first term of the last line is the treatment effect of having been treated for *two periods*. The second and third terms combined capture the deviation in time trends between fully-treated and untreated potential outcomes. The coefficient at t^* :

For $j = t^*$:

$$\begin{aligned}
\gamma_j^{pre} &= \mathbb{E}[Y_{t^*}(\vec{0}_M, 1) - Y_{t^*+1}(\vec{0}_{M-1}, 1, 1) \mid S = s] - \mathbb{E}[Y_{t^*}(\vec{1}_M, 1) - Y_{t^*+1}(\vec{1}_M, 1) \mid S = 0] \\
&= \underbrace{\mathbb{E}[Y_{t^*+1}(\vec{0}_{M-1}, 0, 1) - Y_{t^*+1}(\vec{0}_{M-1}, 1, 1) \mid S = s]}_{\text{(negative) Lingering effect of one prior treated time period}} \\
&\quad + \underbrace{\mathbb{E}[Y_{t^*}(\vec{0}_M, 1) - Y_{t^*+1}(\vec{0}_M, 1) \mid S = s]}_{\text{Time trend for first-treated potential outcomes from } t^* \text{ to } t^*+1} \\
&\quad - \underbrace{\mathbb{E}[Y_{t^*}(\vec{1}_M, 1) - Y_{t^*+1}(\vec{1}_M, 1) \mid S = 0]}_{\text{Time trend for fully treated potential outcomes from } t^* \text{ to } t^*+1}
\end{aligned}$$

This decomposition is symmetric to the previous time periods, and holds by adding and subtracting $\mathbb{E}[Y_{t^*+1}(\vec{0}_{M-1}, 0, 1) \mid S = s]$. If lingering treatment effects are stable over time (Assumption 5(a)), the last two terms in this decomposition cancel. However, the remaining term indicates that even when the time trends cancel each other out, the coefficient on γ_{t^*} recovers the difference, at $t^* + 1$, between potential outcomes from becoming treated at t^* and becoming treated one period later. When treatment effects carry over for $M > 1$ time periods, this lingering effect is ambiguous. $\square$

This decomposition exercise demonstrates that a regression approach to the event study estimation method is sensitive to the specified reference period, especially when treatment effects carry over for more than one period. Additionally, the decompositions also show that, even when correctly specified for the target treatment effect, event study coefficients recover the *negative* of the treatment effect, which has significant implications for qualitative interpretation of the coefficients and should be taken into consideration accordingly.

Consider the event study regression normalized to t^* ,

$$Y_{it} = \beta_1 S_i + \lambda_t + \sum_{j=1}^{t^*-1} \gamma_j^{pre} S_i \cdot \mathbf{1}(t = j) + \sum_{j=t^*+1}^T \gamma_j^{post} S_i \cdot \mathbf{1}(t = j) + \epsilon_{it}. \quad (31)$$

As shown in Proposition A.1, the γ_j^{pre} coefficients recover the negative of $\text{EMATE}(s, j)$ for each pre-switch period, j . An additional nice property of this specification is that it can be used to perform an over-identification test. Under Mean Outcome Convergence, the research design requires not only common time-trends but also common level outcomes across groups. Therefore, in this setting γ_j^{post} coefficients and β_1 are jointly informative about the design. Because of this feature of the research design, the event study regression is over-identified.

Proposition A.3 *In the event study regression in 31, under Mean Outcome Convergence and Assumptions 1-4, for all $j \in [t^* + M, T]$, $\gamma_j^{post} = -\beta_1$.*

Proof. Under Mean Outcome Convergence, and with one carryover period, the estimated γ_j^{post} coefficients decompose to:

$$\begin{aligned} \hat{\gamma}_j^{post} &= \frac{\sum_{i=1}^n (Y_{ij} - Y_{it^*}) \cdot S_i}{\sum_{i=1}^n S_i} - \frac{\sum_{i=1}^n (Y_{ij} - Y_{it^*}) \cdot (1 - S_i)}{\sum_{i=1}^n (1 - S_i)} \\ &= \frac{\sum_{i=1}^n (Y_{ij}) \cdot S_i}{\sum_{i=1}^n S_i} - \frac{\sum_{i=1}^n (Y_{ij}) \cdot (1 - S_i)}{\sum_{i=1}^n (1 - S_i)} - \left(\frac{\sum_{i=1}^n (Y_{it^*}) \cdot S_i}{\sum_{i=1}^n S_i} - \frac{\sum_{i=1}^n (Y_{it^*}) \cdot (1 - S_i)}{\sum_{i=1}^n (1 - S_i)} \right) \end{aligned}$$

Under the research design, average outcomes for all time periods $j \geq t^* + M$ are mean independent of group at the population level. Writing terms as sample means,

$$\hat{\gamma}_j^{post} = \bar{Y}_j^1 - \bar{Y}_j^0 - \left(\bar{Y}_{t^*}^1 - \bar{Y}_{t^*}^0 \right) \quad (32)$$

Taken together, the first two terms are the difference in average outcomes between Switchers and Stayers in post-switch period j . Under Mean Outcome Convergence, the population averages are equal, and by the central limit theorem, the sample mean difference is a draw from a normal distribution with mean 0 and variance Σ_j . By Slutsky's theorem, $\hat{\gamma}_j^{post} \xrightarrow{d} \mathcal{N}\left(\mathbb{E}[Y_{t^*} | S = 1] - \mathbb{E}[Y_{t^*} | S = 0], \mathbf{V}\right)$. It is clear that if there are level differences in outcomes at t^* , then γ_j^{post} carries these differences, and the coefficient will be nonzero, as will its unbiased estimate. A “post-trends test” that tests for nonzero coefficients (symmetric to a pre-trends test) will not correctly assess the design.

Notice that under the research design, the asymptotic mean of $\hat{\gamma}_j^{post}$ is the difference in average outcomes between groups at time t^* , and is invariant in j . This motivates a joint hypothesis test of the coefficients holding equal at some non-zero value:

$$H_0 : \text{For all } j \geq t^* + M, \quad \gamma_j^{post} = \gamma_{j+1}^{post}$$

The other regression coefficients are also informative about the reserch design. To demonstrate this, I next perform a Frisch-Waugh-Lovell decomposition of β_1 .

Let $\mathbf{X}_{it} = \{\alpha_t, S_i \cdot \mathbf{1}(t = j \neq t^*)\}$ be the vector of regressors excluding the one corresponding to the β_1 coefficient for unit i at time t . α_t is a vector of indicator functions: $\alpha_t = \{\mathbf{1}(t = 1), \mathbf{1}(t = 2), \dots, \mathbf{1}(t = T)\}$ that creates time fixed effects. $S_i \cdot \mathbf{1}(t = j \neq t^*)$ are interaction terms between the cohort indicator and the time period.

First, I project S_i on $\mathbf{X}_{it}$:

$$S_{it} = \alpha_t + \sum_{j \neq t^*} \lambda_j S_i \cdot \mathbf{1}(t = j) + u_{it}$$

For all $t \neq t^*$, the $\mathbf{X}_{it}$ vector contains the term S_i which is also the outcome of the regression. The interaction term perfectly predicts the outcome for these time periods, the α_t coefficients are zero and the residual is zero. When $t = t^*$, the interaction term is zero, and the only regressor is the time intercept, α_{t^*} . This means that $\hat{\alpha}_{t^*} = \bar{S}$. The fitted values are $\hat{S}_{it} = S_i$ for $t \neq t^*$ and $\hat{S}_{it} = \bar{S}$ for $t = t^*$. Both are summarized in the expression for fitted values below:

$$\hat{S}_{it} = S_i \cdot \mathbf{1}(t \neq t^*) + \bar{S} \cdot \mathbf{1}(t = t^*)$$

The residuals from this projection are:

$$\begin{aligned} \hat{u}_{it} &= S_i - \hat{S}_{it} \\ &= S_i - S_i \cdot \mathbf{1}(t \neq t^*) + \bar{S} \cdot \mathbf{1}(t = t^*) \\ &= (S_i - \bar{S}) \cdot \mathbf{1}(t = t^*) \end{aligned}$$

As noted previously, these residuals are zero for all $t \neq t^*$ because $\lambda_t = 1$. Since the residual from the projection of S_i on $\mathbf{X}_{it}$ is only non-zero for $t = t^*$, I next project Y_{it^*} on $\mathbf{X}_{it^*}$

$$\begin{aligned} Y_{it^*} &= \alpha_{t^*} + \sum_{j \neq t^*} \gamma_j S_i \cdot \mathbf{1}(t^* = j) + v_{it^*} \\ Y_{it^*} &= \alpha_{t^*} + v_{it^*} \end{aligned}$$

For $t = t^*$, all of the interaction terms are zero, and the only regressor is the time intercept. The projection is therefore just the mean of Y_{t^*} . The fitted values are:

$$\hat{Y}_{it^*} = \bar{Y}_{t^*}$$

The residuals from this projection are:

$$\hat{v}_{it} = Y_{it^*} - \hat{Y}_{it^*} = Y_{it^*} - \bar{Y}_{t^*}$$

By the FWL theorem,

$$\begin{aligned}
\hat{\beta}_1 &= \frac{\sum_i \sum_t \widehat{u}_{it} \cdot \widehat{v}_{it}}{\sum_i \sum_t \widehat{u}_{it}^2} = \frac{\sum_i \sum_t (S_i - \bar{S}) \cdot \mathbf{1}(t = t^*) \cdot \widehat{v}_{it}}{\sum_i \sum_t ((S_i - \bar{S}) \cdot \mathbf{1}(t = t^*))^2} \\
&= \frac{\sum_i (S_i - \bar{S}) \cdot (Y_{it^*} - \bar{Y}_{t^*})}{\sum_i (S_i - \bar{S})^2}
\end{aligned} \tag{33}$$

The first equality is the FWL theorem, the second equality holds by substituting in the residuals from the first projection, and the third equality follows because $\widehat{u}_{it} = 0$ for all $t \neq t^*$.

The numerator:

$$\begin{aligned}
\sum_i (S_i - \bar{S}) \cdot (Y_{it^*} - \bar{Y}_{t^*}) &= \sum_i \left(S_i Y_{it^*} - \bar{S} \bar{Y}_{t^*} \right) \\
&= n \cdot \hat{p} \bar{Y}_{t^*}^1 - n \hat{p} \cdot \bar{Y}_{t^*} \\
&= n \hat{p} \left(\bar{Y}_{t^*}^1 - (\hat{p} \bar{Y}_{t^*}^1 + (1 - \hat{p}) \bar{Y}_{t^*}^0) \right) \\
&= n \hat{p} (1 - \hat{p}) (\bar{Y}_{t^*}^1 - \bar{Y}_{t^*}^0)
\end{aligned} \tag{34}$$

The first equality follows by expanding the term inside the sum, and replaces $\bar{S} = \hat{p}$ and $\sum_i Y_{it^*} = \bar{Y}_{t^*}$. The second holds because $\sum_i S_i Y_{it^*} = n \hat{p} \bar{Y}_{t^*}^1$ and $\sum_i \bar{S} \bar{Y}_{t^*} = n \hat{p} \bar{Y}_{t^*}$. Line 3 expands the unconditional sample average of Y_{t^*} into a weighted average of the group-specific averages. With some algebra, the last line expresses the numerator of the expression in 33 as the average difference in outcomes between the two groups at t^* weighted by the asymptotic variance of $\hat{p}$.

The denominator:

$$\begin{aligned}
\sum_i (S_i - \bar{S})^2 &= \sum_i S_i - 2\hat{p} \sum_i S_i + n\hat{p}^2 \\
&= n\hat{p} + n\hat{p}^2 \\
&= n\hat{p}(1 - \hat{p})
\end{aligned} \tag{35}$$

The first equality holds by expanding the squared term, substituting $\bar{S} = \hat{p}$, and because when S_i is binary, $S_i^2 = S_i$. The second line follows by combining like terms and because $\sum_i S_i = n\hat{p}$. The last expression states that the denominator of 33 is the asymptotic variance of $\hat{p}$.

Combining the expressions in 34 and 35, the FWL decomposition of $\hat{\beta}_1$ in 33 is:

$$\hat{\beta}_1 = \bar{Y}_{t^*}^1 - \bar{Y}_{t^*}^0$$

which is the difference in mean outcomes across groups at time t^* . By the Central Limit Theorem, $\hat{\beta}_1 \xrightarrow{p} \mathbb{E}[Y_{t^*} \mid S = 1] - \mathbb{E}[Y_{t^*} \mid S = 0]$. Under Mean Outcome Convergence,

$$\begin{aligned}\mathbb{E}[\widehat{\gamma}_j^{post}] &= [Y_{t^*} \mid S = 0] - \mathbb{E}[Y_{t^*} \mid S = 1] \\ &= -\beta_1 \quad \text{for all } j \geq t^* + M.\end{aligned}$$

□

Equality of the population coefficients holds mechanically under the research design, and statistical tests conducted on the sample estimates from the data can provide empirical support for the design. These statistical tests must consider the inflated type-1 error rate of multiple hypothesis testing as well as the within-group covariance in mean outcomes over time. For methods equipped to handle these concerns, see Callaway and Sant’Anna (2021) and related literature on simultaneous multiple hypothesis testing.

While the correctly specified event study regression can be used to recover treatment effects and provide support for the research design, the simple estimation method in section 4 is preferable because it is adaptable to various methods of testing the research design, yields directly interpretable coefficients (as opposed to negative coefficients in the regression approach), and can easily be written as an influence function to adopt advanced covariate balancing techniques such as Doubly Robust estimation and machine learning.

B Inverse Propensity Score Weighting Estimator

Suppose that Conditional Mean Outcome Convergence holds as in 25 for arbitrary, continuous X .

Proposition B.1: *Under Assumptions 1-4, and with Conditional Mean Outcome Convergence:*

$$\begin{aligned}\mathbb{E}[Y_t(\vec{1}_{t^*+M:T}) \mid X] &\perp S \\ 0 \leq \Pr(S \mid X = x) &\leq 1 \quad \text{for all } S, \text{ for all } x \in \mathcal{X}\end{aligned}$$

the IPW estimator for $TATE(s, t)$ is identified for all $t \leq t^* - 1$ as:

$$\widehat{TATE}^{IPW}(s, t) = \frac{1}{n} \sum_{i=1}^n \left(\frac{\hat{p}(X_i)(1 - S_i)}{\hat{p}(1 - \hat{p}(X_i))} - \frac{S_i}{\hat{p}} \right) Y_{i,t}$$

Proof. For the derivation below, let $p(x) = \Pr(S = 1 \mid X = x)$, and let $p = \Pr(S = 1)$. For simplicity, I also assume that $M = 1$

$$\begin{aligned}TATE(s, t) &= \mathbb{E}[Y_t(\vec{1}_{M+1}) \mid S = s] - \mathbb{E}[Y_t(\vec{0}_{M+1}) \mid S = s] \\ &= \mathbb{E}[\mathbb{E}[Y_t(\vec{1}_{M+1}) \mid S = 0, X = x] \mid S = s] - \mathbb{E}[Y_t \mid S = s]\end{aligned} \tag{36}$$

The first equality is the definition of the target parameter, and the second equality holds by the

law of iterated expectations and Conditional Mean Outcome Convergence. Focusing on the first term,

$$\begin{aligned}
\mathbb{E}[\mathbb{E}[Y_t(\vec{1}_{M+1}) \mid S = 0, X = x] \mid S = s] &= \mathbb{E}\left[\frac{f_{X|S}(x \mid S = s)}{Pr(X = x)} \mathbb{E}[Y_t(\vec{1}_{M+1}) \mid S = 0, X = x]\right] \\
&= \mathbb{E}\left[\frac{f_{S|X}(S = s \mid x)}{Pr(S = s)} \mathbb{E}[Y_t(\vec{1}_{M+1}) \mid S = 0, X = x]\right] \\
&= \mathbb{E}\left[\frac{p(x)}{p} \mathbb{E}[Y_t(\vec{1}_{M+1}) \mid S = 0, X = x]\right] \\
&= \mathbb{E}\left[\frac{1-p}{1-p(x)} \frac{p(x)}{p} \mathbb{E}[Y_t(\vec{1}_{M+1}) \mid S = 0, X = x] \mid S = 0\right] \\
&= \mathbb{E}\left[\frac{p(x)(1-p)}{p(1-p(x))} Y_t(\vec{1}_{M+1}) \mid S = 0\right] \\
&= \mathbb{E}\left[\frac{p(x)(1-p)}{p(1-p(x))} Y_t \mid S = 0\right].
\end{aligned}$$

The first equality holds by expanding the conditional probability. The second equality holds by Bayes' rule and the third expression follows by substituting $p(x) = Pr(S = 1 \mid X = x)$ and $p = Pr(S = 1)$. The fourth equality follows because $\mathbb{E}[g(x) \mid S = 0] = \mathbb{E}[\frac{1-p(x)}{1-p} g(x)]$. The fifth line applies the law of iterated expectations, and the equality with the expression in the sixth line holds because for all pre-switch periods, $\mathbb{E}[Y_t(\vec{1}_{M+1})]$ is observed for the Switchers. Combining this with the last expression in 36,

$$\begin{aligned}
TATE(s, t) &= \mathbb{E}\left[\frac{p(x)(1-p)}{p(1-p(x))} Y_t \mid S = 0\right] - \mathbb{E}[Y_t \mid S = s] \\
&= \mathbb{E}\left[\frac{p(x)(1-p)}{p(1-p(x))} \frac{1}{1-p} (1-S) Y_t\right] - \mathbb{E}\left[\frac{1}{p} S Y_t\right] \\
&= \mathbb{E}\left[\frac{p(x)(1-S)}{p(1-p(x))} Y_t\right] - \mathbb{E}\left[\frac{S}{p} Y_t\right] \\
&= \mathbb{E}\left[\frac{p(x)(1-S)}{p(1-p(x))} Y_t - \frac{S}{p} Y_t\right]
\end{aligned} \tag{37}$$

The first line substitutes the derived expression for $\mathbb{E}[\mathbb{E}[Y_t(\vec{1}_{M+1}) \mid S = 0, X = x] \mid S = s]$. The second line holds because $\mathbb{E}[Y_t \mid S = 0] = \frac{1}{1-p} \mathbb{E}[(1-s)Y_t]$ and symmetrically, $\mathbb{E}[Y_t \mid S = 1] = \frac{1}{p} \mathbb{E}[S Y_t]$. The third and fourth lines simplify terms and pass the unconditional expectation through linearly. Using inverse propensity score weighting (IPW), the TATE is identified in the last line of the above derivation. It follows that estimation of the treatment effect is the sample analog of this line:

$$\widehat{TATE}(s, t) = \frac{1}{n} \sum_{i=1}^n \left(\frac{\hat{p}(X_i)(1-S_i)}{\hat{p}(1-\hat{p}(X_i))} - \frac{S_i}{\hat{p}} \right) Y_{i,t}$$

□

The propensity score weighting approach is preferable to methods such as regression adjustment because a key feature of this paper is that potential outcomes models can be nonlinear and heterogeneous across units. Because the term that is re-weighted corresponds to treated potential outcomes, which are dynamic in event time, regression adjustment that assumes a linear functional form is not appropriate. The IPW approach is also more robust in this setting because Mean Outcome Convergence conditional on covariates is testable, meaning the likelihood of cohort membership conditional on X is observable while the functional form assumption for regression adjustment is fundamentally untestable.

Note that the IPW approach requires an additional overlap condition: For all $x \in \mathcal{X}$, $0 < Pr(S = 1 | X = x) < 1$. This condition is necessary to ensure that the propensity score is well-defined and that there is sufficient variation in treatment assignment across the covariate space. If this overlap condition cannot be satisfied, it might be necessary to restrict analysis to a subset of the units. Furthermore, if $Pr(S = 1 | X = x)$ is very small or very large, the estimator suffers from weak overlap. See Dorn (2025) and Ma et al. (2023) for applicable methods that address this in panel data models. Additionally, it is worth mentioning that the IPW approach does not improve efficiency when Mean Outcome Convergence holds unconditionally, and should only be used when the research design requires stronger conditions than those derived in the main text.

C Application: Supplementary Figures and Tables

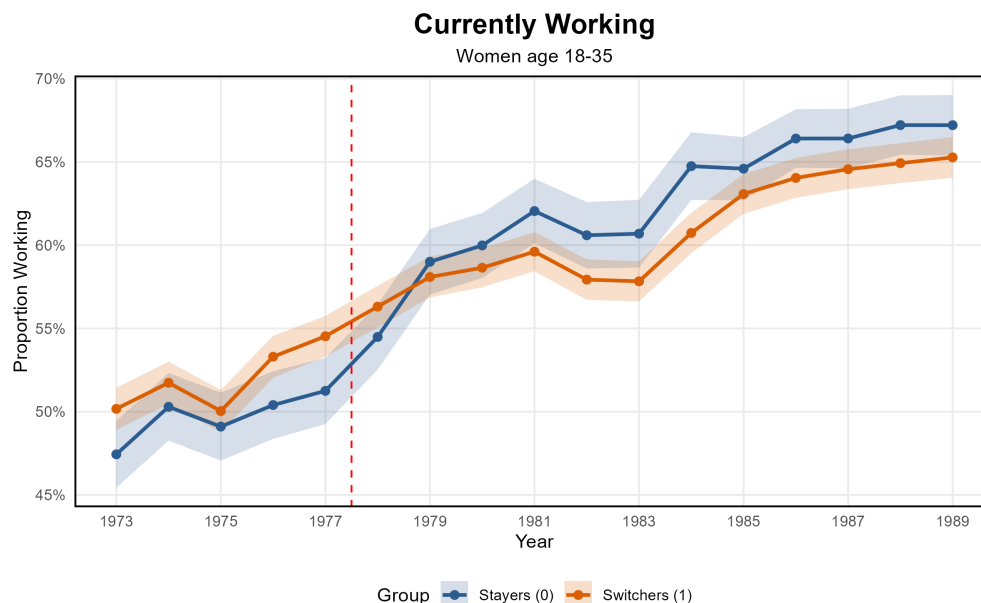


Figure C.1: Employment status among all women, aged 18-35 in Switcher and Stayer states. Shaded regions represent 95% confidence intervals using ASEC individual survey weights.

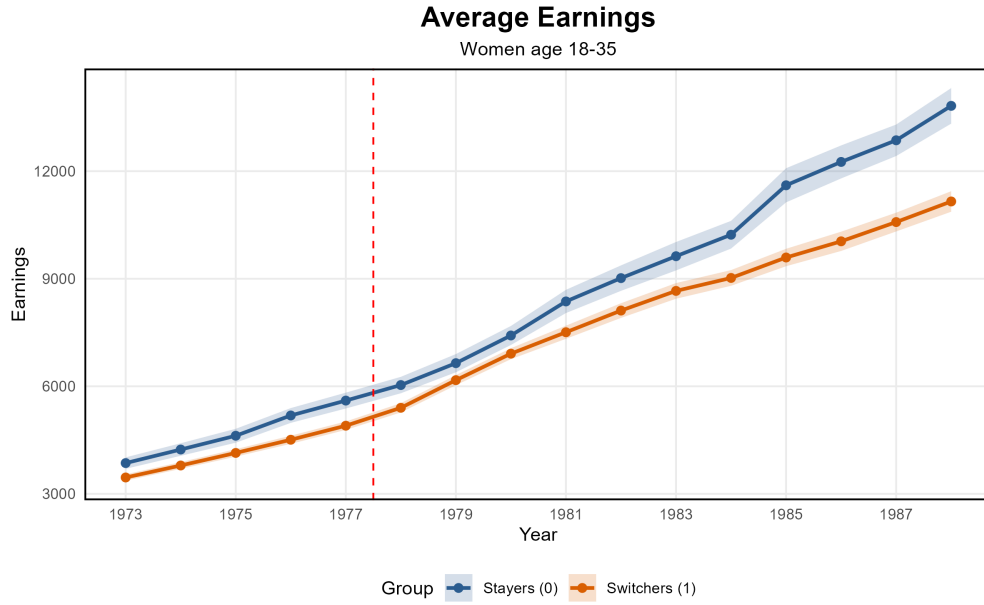


Figure C.2: Earnings among women aged 18-35 with gainful employment in Switcher and Stayer states. Shaded regions represent 95% confidence intervals using ASEC individual survey weights.

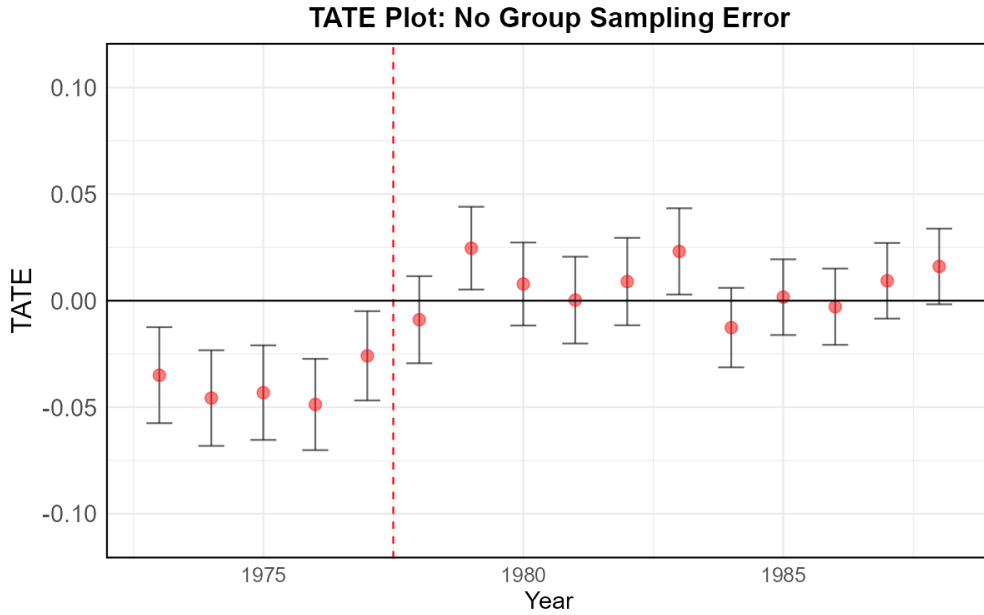


Figure C.3: TATE(s,t) coefficients for Switchers at time t . Bars around each point estimate represent 95% confidence intervals with standard errors computed based on the estimation method in Section 4. Confidence intervals are not adjusted for sampling error from $\hat{p}$. All point estimates and standard errors are computed using ASEC individual survey weights.

Table C.1: TATE(s,t) Point Estimates for Switchers

Year	Estimated TATE	Standard Error
1973	-0.035	0.017
1974	-0.046	0.017
1975	-0.043	0.016
1976	-0.049	0.016
1977	-0.026	0.016
1978	-0.009	0.017
1979	0.025	0.017
1980	0.008	0.017
1981	0.000	0.017
1982	0.009	0.017
1983	0.023	0.017
1984	-0.013	0.016
1985	0.002	0.015
1986	-0.003	0.016
1987	0.009	0.016
1988	0.016	0.016

Estimated Total Average Treatment Effect for Switchers by year. Standard errors are computed using the variance estimator derived in Section 2 of the main text. All estimates are computed using individual-level ASEC weights.

Table C.2: Multiplier bootstrap: uniform confidence band for difference in means

Year	$\hat{\delta}$	Standard Error (Bootstrap)	95% Uniform Confidence Band
1978	-0.007	0.010	[-0.024, 0.010]
1979	0.026	0.010	[0.008, 0.043]
1980	0.013	0.011	[-0.005, 0.032]
1981	0.001	0.010	[-0.016, 0.019]
1982	0.011	0.011	[-0.008, 0.029]
1983	0.025	0.010	[0.008, 0.042]
1984	-0.007	0.009	[-0.024, 0.009]
1985	-0.003	0.009	[-0.019, 0.013]
1986	-0.004	0.010	[-0.020, 0.013]
1987	0.005	0.009	[-0.011, 0.020]
1988	0.014	0.009	[-0.002, 0.029]

Estimated difference in mean outcomes between Switchers and Stayers in all post-switch periods are reported along with multiplier bootstraps algorithm standard errors and 95% Confidence Bands.